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(54) **CONFIGURATION FOR OLEFINS  
PRODUCTION**

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**B01D 3/06** (2006.01)

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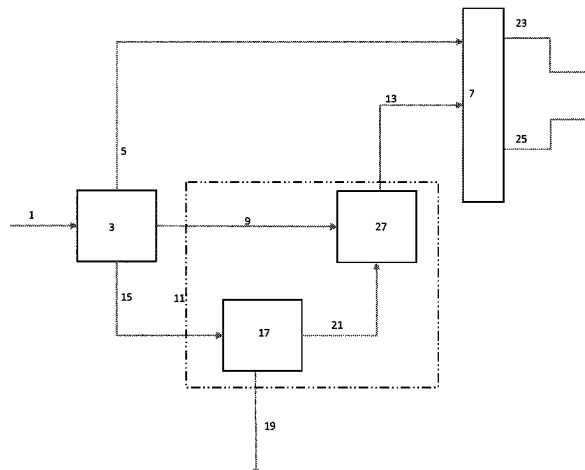
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**ABSTRACT**

Processes herein may be used to thermally crack various hydrocarbon feeds, and may eliminate the refinery altogether while making the crude to chemicals process very flexible in terms of crude. In embodiments herein, crude is progressively separated into at least light and heavy fractions. Depending on the quality of the light and heavy fractions, these are routed to one of three upgrading operations, including a fixed bed hydroconversion unit, a fluidized catalytic conversion unit, or a residue hydrocracking unit that may utilize an ebullated bed reactor. Products from the upgrading operations may be used as feed to a steam cracker.

**18 Claims, 4 Drawing Sheets****Related U.S. Application Data**

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**65/12** (2013.01); **C10L 1/08** (2013.01); **B01D 3/06** (2013.01); **B01D 19/0057** (2013.01); **B01J 2219/0004** (2013.01); **C10G 2300/1044** (2013.01); **C10G 2300/1048** (2013.01); **C10G 2300/202** (2013.01); **C10G 2300/205** (2013.01); **C10G 2300/206** (2013.01); **C10G 2300/301** (2013.01); **C10G 2300/302** (2013.01); **C10G 2300/308** (2013.01); **C10G 2300/40** (2013.01); **C10G 2300/4006** (2013.01); **C10G 2300/4081** (2013.01); **C10G 2300/807** (2013.01); **C10G 2400/20** (2013.01); **C10G 2400/22** (2013.01); **C10G 2400/30** (2013.01)

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See application file for complete search history.

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Figure 1

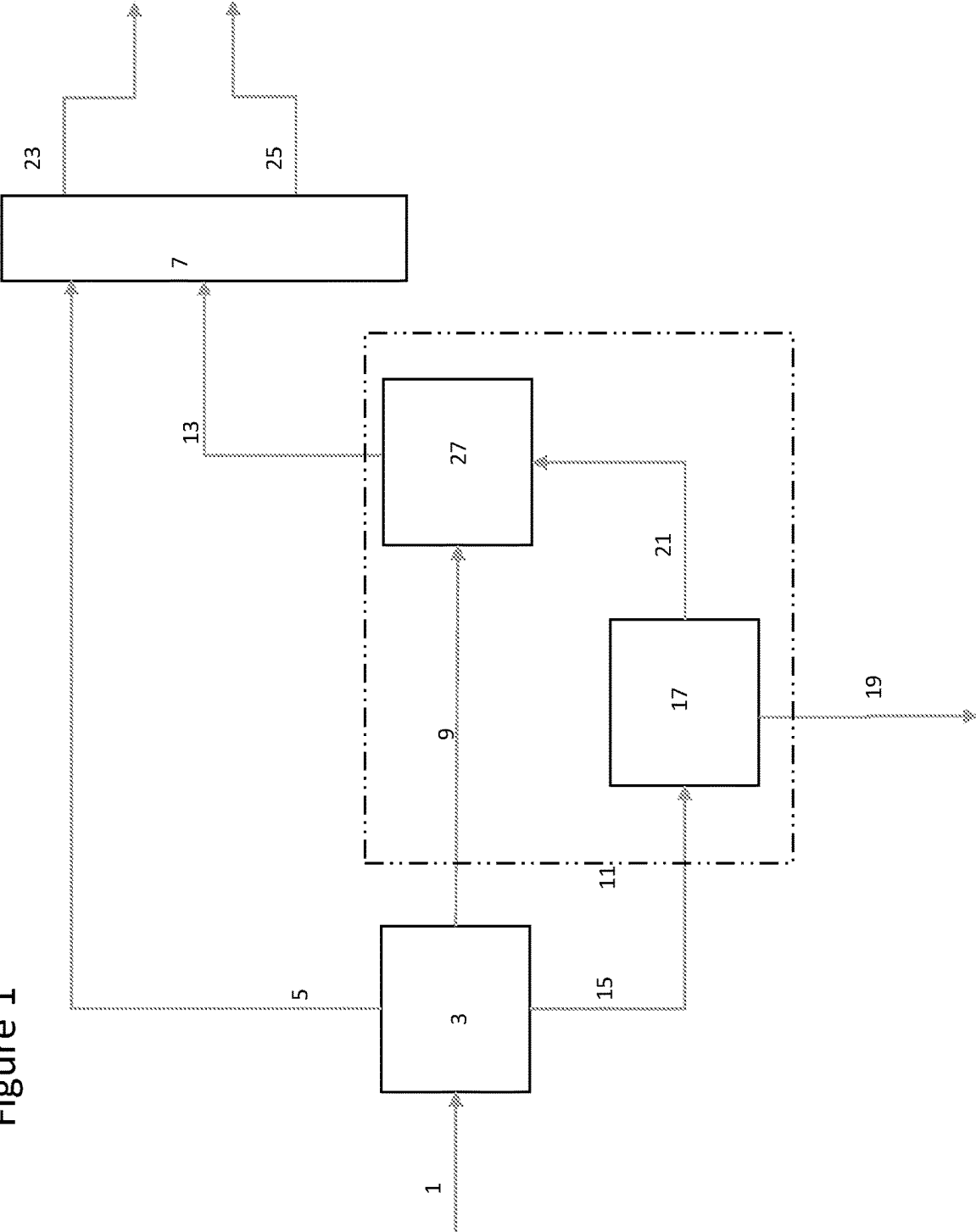


Figure 2

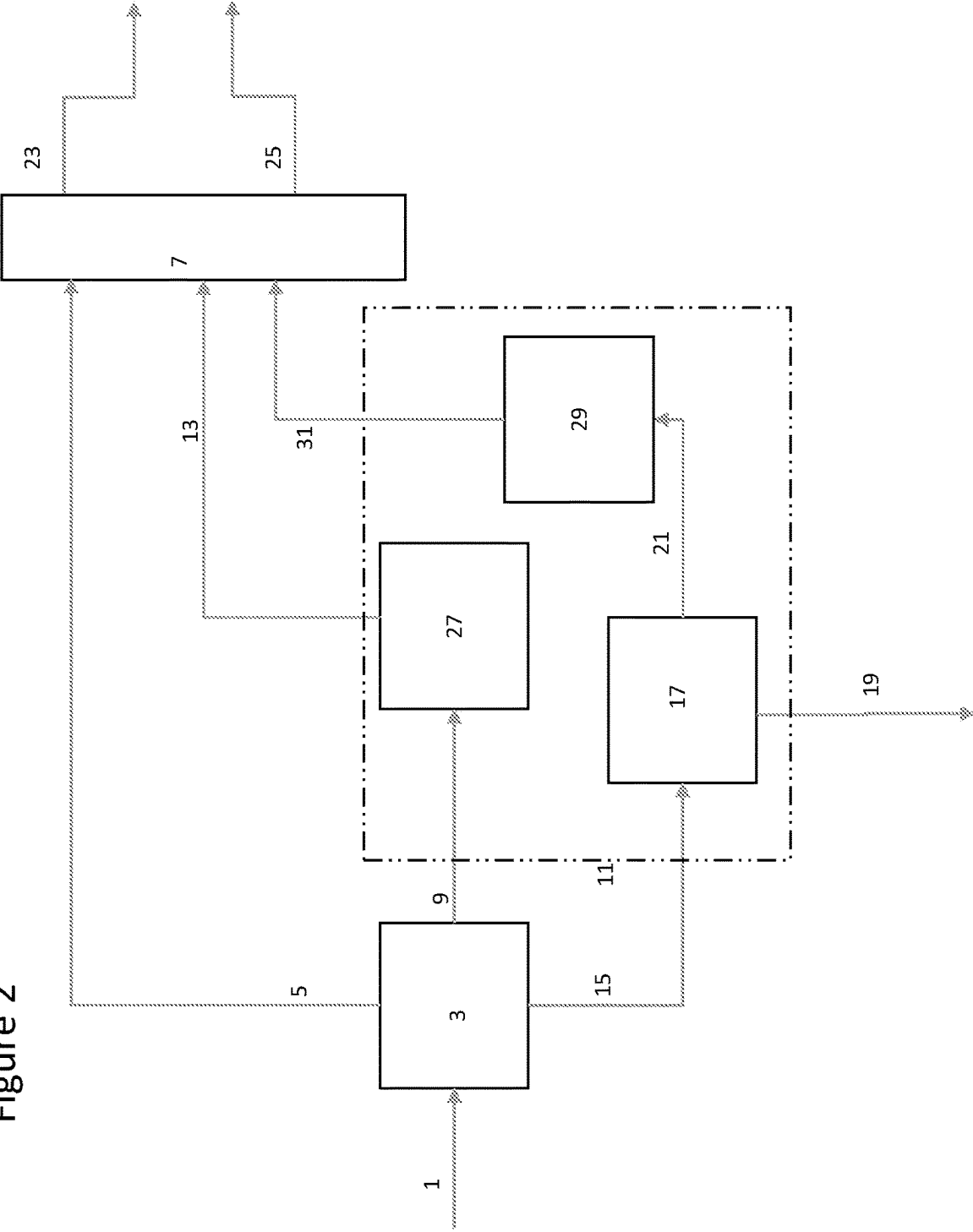


Figure 3

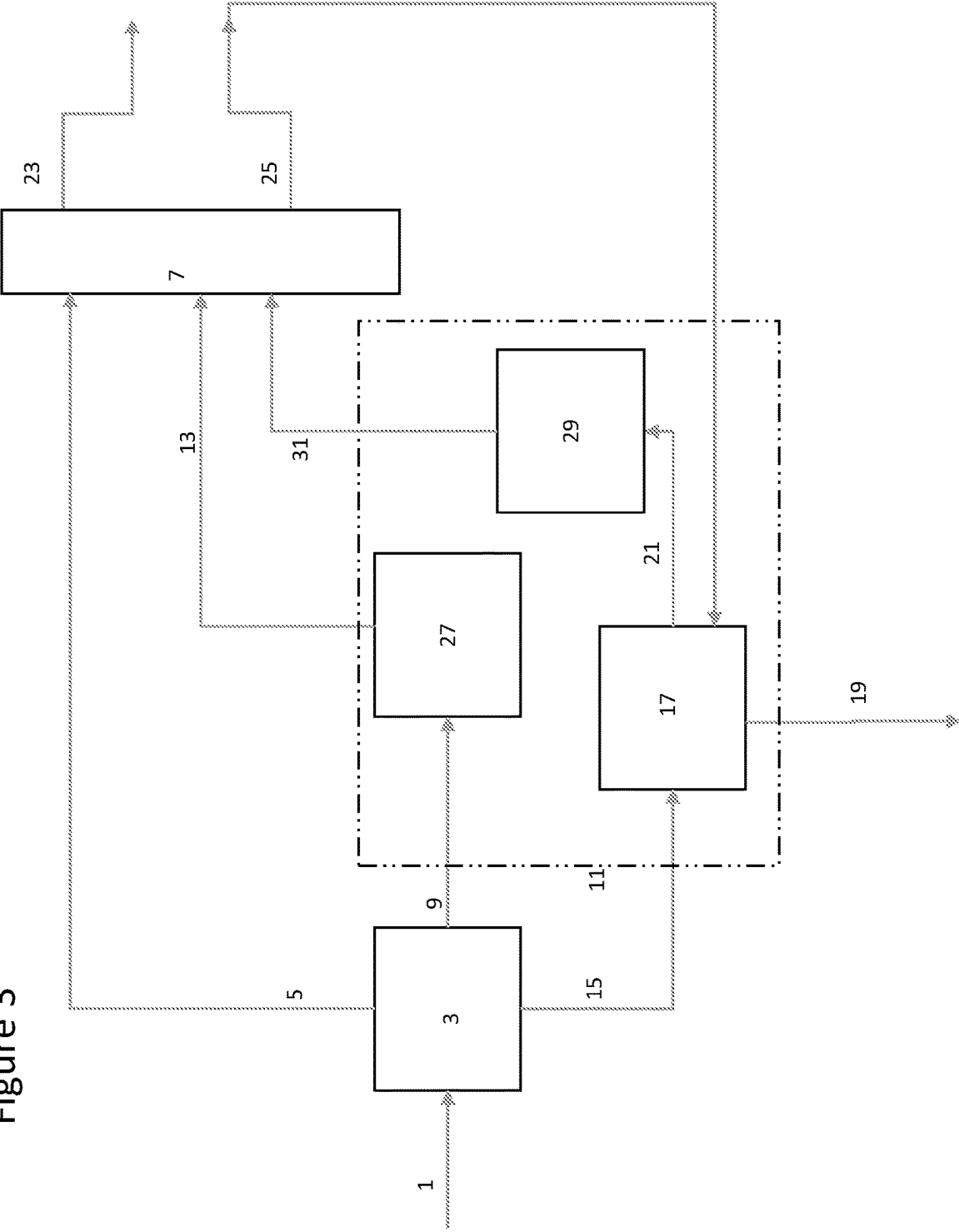
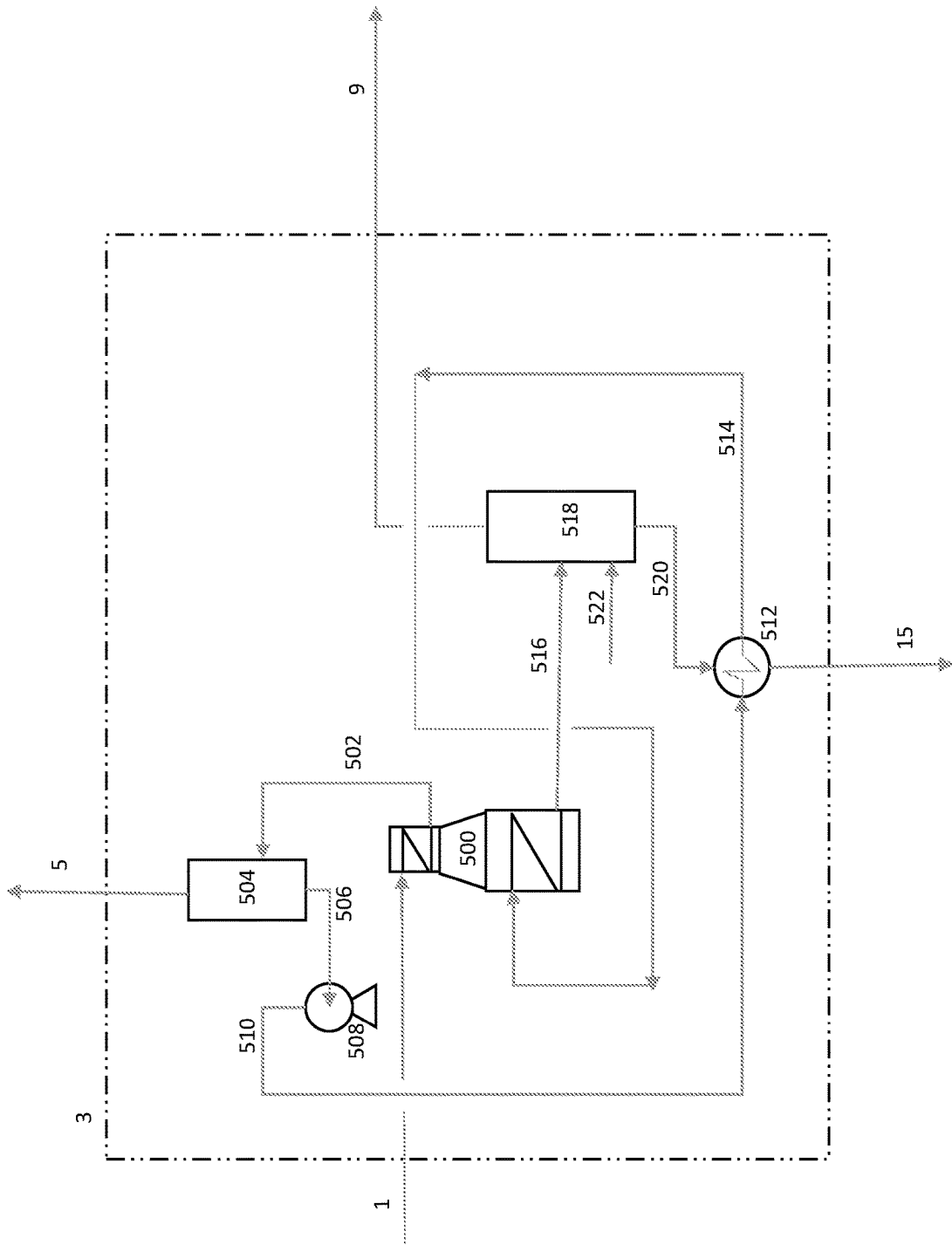


Figure 4



# CONFIGURATION FOR OLEFINS PRODUCTION

## CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a divisional of U.S. patent application Ser. No. 16/820,038, now U.S. Pat. No. 11,180,706, filed on Mar. 16, 2020, and pursuant to 35 U.S.C. § 119(e), claims priority to U.S. Provisional Application Ser. Nos. 62/819,270, 62/819,282, 62/819,247, 62/819,229, and 62/819,315, each filed Mar. 15, 2019, and each of which is herein incorporated by reference in its entirety.

## FIELD OF THE DISCLOSURE

Embodiments herein relate to processes and systems for producing petrochemicals, such as olefins and aromatics, from crude oil and low value heavy hydrocarbon streams.

## BACKGROUND

High-boiling compounds in crude oil may cause significant operational issues if they are sent to a steam cracker. High boiling compounds have a propensity to form coke, due in large part to their high asphaltene content. Therefore, the high boiling compounds are typically removed before sending the lighter fractions to different petrochemicals units, such as a steam cracker or an aromatic complex. The removal process, however, increases the capital cost of the overall process and lowers profitability, as the removed high-boiling compounds can only be sold as low-value fuel oil. In addition, conversion of vacuum residue without significant formation of HPNAs (heavy polynuclear aromatics) that are detrimental to steam cracker furnaces downstream of the process has been a challenge to date.

U.S. Pat. No. 3,617,493 describes a process in which crude oil is sent to the convection section of a steam cracker and then to a separation zone, where the portion of the feed boiling below about 450° F. is separated from the rest of the feed and then sent, with steam, into the high temperature portion of the steam cracker and subjected to cracking conditions.

U.S. Pat. No. 4,133,777 teaches a process in which feed oil initially flows downwardly in trickle flow through a fixed bed of HDM catalysts, and then passes downwardly through a fixed bed of promoted catalysts containing selected GROUP VI and GROUP VIII metals, with very little hydrocracking occurring in this combination process.

U.S. Pat. No. 5,603,824 disclosed a process of upgrading a waxy hydrocarbon feed mixture containing sulfur compounds which boils in the distillate range, in order to reduce sulfur content and 85% point while preserving the high octane of naphtha by-products and maximizing distillate yield. The process employs a single, downflow reactor having at least two catalyst beds and an inter-bed redistributor between the beds. The top bed contains a hydrocracking catalyst, preferably zeolite beta, and the bottom bed contains a dewaxing catalyst, preferably ZSM-5.

U.S. Pat. No. 3,730,879 discloses a two-bed catalytic process for the hydrodesulfurization of crude oil or a reduced fraction, in which at least 50 percent of the total pore volume of the first-bed catalytic consists of pores in the 100-200 Angstrom diameter range.

U.S. Pat. No. 3,830,720 discloses a two-bed catalytic process for hydrocracking and hydrodesulfurizing residual oils, in which a small-pore catalyst is disposed upstream of a large-pore catalyst.

U.S. Pat. No. 3,876,523 describes a novel catalyst and a process for catalytically demetalizing and desulfurizing hydrocarbon oils comprising residual fractions. The process described therein utilizes a catalyst comprising a hydrogenation component, such as cobalt and molybdenum oxides, composited on an alumina. Although this catalyst is highly effective for demetalization of residual fractions and has good stability with time on stream, its utility is remarkably improved when this catalyst is employed in a particular manner in combination with a second catalyst having different critical characteristics. A catalyst of the type described in U.S. Pat. No. 3,876,523 will be referred as a first catalyst, it being understood that this first catalyst is to be situated upstream of the second catalyst having different characteristics.

U.S. Pat. No. 4,153,539 discloses that improved hydrogen utilization and/or higher conversions of desired product is obtained in hydrotreating or hydrocracking processes when using amphora particles for hydrotreating of light hydrocarbon fractions, catalytic reforming, fixed-bed alkylation processes, and the like.

U.S. Pat. No. 4,016,067 discloses that hydrocarbon oils, preferably residual fractions, are catalytically hydroprocessed to very effectively remove both metals and sulfur and with particularly slow aging of the catalyst system by contacting the oil sequentially with two catalysts of different characteristics. The first catalyst, located upstream of the second catalyst, is characterized by having at least 60% of its pore volume in pores greater than 100 Å. in diameter and other characteristics hereinafter specified. The second catalyst, located downstream with respect to the first catalyst, is characterized by having a major fraction of its pore volume in pores less than 100 Å. in diameter.

The dual catalyst apparatus of U.S. Pat. No. 4,016,067 is used to demetallize and/or desulfurize any hydrocarbon oil that has metals and/or sulfur content-undesirably high for a particular application. The dual catalyst apparatus is particularly effective for preparing low metals and/or low sulfur content feedstocks for catalytic cracking or for coking. In the processing to remove metals and sulfur, and hydrocarbon oil also is concomitantly enriched in hydrogen, making it an even more suitable chargestock for either of these processes.

In general, these and other prior processes for converting whole crudes typically convert less than 50 percent of the crude to the more desirable end products, including petrochemicals such as ethylene, propylene, butenes, pentenes, and light aromatics, for example. Generally, 20 percent of the whole crude is removed up front in processing, removing the heaviest components that are hard to convert. About another 20 percent of the whole crude is typically converted to pyrolysis oil, and about 10 percent is over-converted to methane.

## SUMMARY

A process for converting whole crudes and other heavy hydrocarbon streams to produce olefins and/or aromatics, the process including: separating a whole crude into at least a light boiling fraction, a medium boiling fraction, and a high boiling residue fraction; separating the high boiling fraction in a solvent deasphalting unit to produce a deasphalted oil fraction and a pitch fraction; destructively hydrogenating the medium boiling fraction and the deasphalted oil fraction to produce a hydrotreated effluent; feeding the hydrotreated effluent and the light boiling fraction to a steam cracker to convert hydrocarbons therein into one or more light olefins and a pyrolysis oil.

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A system for converting whole crudes and other heavy hydrocarbon streams to produce olefins, the system including: a separation unit for separating a whole crude into at least a light boiling fraction, a medium boiling fraction, and a high boiling residue fraction; a solvent deasphalting unit for deasphalting the high boiling residue fraction and producing a deasphalted oil fraction and a pitch fraction; a conditioning system for destructively hydrogenating the medium boiling fraction and the deasphalted fraction to produce a hydrotreated effluent; a steam cracker unit for converting the hydrotreated effluent and the light boiling fraction into one or more light olefins and a pyrolysis oil.

Other aspects and advantages will be apparent from the following description and the appended claims.

### BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a simplified process flow diagram of a system for converting whole crudes and heavy hydrocarbons according to embodiments herein.

FIG. 2 is a simplified process flow diagram of a system for converting whole crudes and heavy hydrocarbons according to embodiments herein.

FIG. 3 is a simplified process flow diagram of a system for converting whole crudes and heavy hydrocarbons according to embodiments herein.

FIG. 4 is a simplified process flow diagram of a system for converting whole crudes and heavy hydrocarbons according to embodiments herein.

### DETAILED DESCRIPTION

As used herein, the term “petrochemicals” refers to hydrocarbons including light olefins and diolefins and C6-C8 aromatics. Petrochemicals thus refers to hydrocarbons including ethylene, propylene, butenes, butadienes, pentenes, pentadienes, as well as benzene, toluene, and xylenes. Referring to a subset of petrochemicals, the term “chemicals,” as used herein, refers to ethylene, propylene, butadiene, 1-butene, isobutylene, benzene, toluene, and para-xylenes.

Hydrotreating is a catalytic process, usually carried out in the presence of free hydrogen, in which the primary purpose when used to process hydrocarbon feedstocks is the removal of various metal contaminants (e.g., arsenic), heteroatoms (e.g., sulfur, nitrogen and oxygen), and aromatics from the feedstock. Generally, in hydrotreating operations cracking of the hydrocarbon molecules (i.e., breaking the larger hydrocarbon molecules into smaller hydrocarbon molecules) is minimized. As used herein, the term “hydrotreating” refers to a refining process whereby a feed stream is reacted with hydrogen gas in the presence of a catalyst to remove impurities such as sulfur, nitrogen, oxygen, and/or metals (e.g. nickel, or vanadium) from the feed stream (e.g. the atmospheric tower bottoms) through reductive processes. Hydrotreating processes may vary substantially depending on the type of feed to a hydrotreater. For example, light feeds (e.g. naphtha) contain very little and few types of impurities, whereas heavy feeds (e.g. ATBs) typically possess many different heavy compounds present in a crude oil. Apart from having heavy compounds, impurities in heavy feeds are more complex and difficult to treat than those present in light feeds. Therefore, hydrotreating of light feeds is generally performed at lower reaction severity, whereas heavy feeds require higher reaction pressures and temperatures.

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Hydrocracking refers to a process in which hydrogenation and dehydrogenation accompanies the cracking/fragmentation of hydrocarbons, e.g., converting heavier hydrocarbons into lighter hydrocarbons, or converting aromatics and/or cycloparaffins(naphthenes) into non-cyclic branched paraffins.

“Conditioning” and like terms as used herein refers to conversion of hydrocarbons by one or both of hydrocracking and hydrotreating. “Destructive hydrogenation” and like terms refers to cracking of the hydrocarbon molecular bonds of a hydrocarbon, and the associated hydrogen saturation of the remaining hydrocarbon fragments, which can create stable lower boiling point hydrocarbon oil products, and may be inclusive of both hydrocracking and hydrotreating.

“API gravity” refers to the gravity of a petroleum feedstock or product relative to water, as determined by ASTM D4052-11.

The integration of conditioning, fractionation, and steam cracking may result in a highly efficient facility, and in some embodiments may convert greater than 55%, greater than 60%, greater than 65%, greater than 70%, greater than 75%, greater than 80%, or greater than 85% of the whole crude to petrochemicals. In other embodiments, the integration of conditioning, fractionation, and steam cracking may result in a highly efficient facility, and in some embodiments may convert greater than 55%, greater than 60%, greater than 65%, greater than 70%, greater than 75%, greater than 80% or greater than 85% of the whole crude to chemicals. Embodiments herein may thus provide systems and processes for conditioning feeds including even the heaviest, most undesirable residuum components into components that can be vaporized and passed into the radiant section of a steam cracker, substantially improving over the low petrochemical conversion of prior processes.

Embodiments herein relate to processes and systems that take crude oil and/or low value heavy hydrocarbons as feed and produces petrochemicals, such as light olefins and diolefins (ethylene, propylene, butadiene, and/or butenes) and aromatics. More specifically, embodiments herein are directed toward methods and systems for making olefins and aromatics by thermal cracking of a pre-conditioned crude oil or condensate. Processes herein may condition the residuum fraction of whole crude oils and natural condensates to produce feedstocks useful as a steam cracker feedstock.

Hydrocarbon mixtures useful in embodiments disclosed herein may include various hydrocarbon mixtures having a boiling point range, where the end boiling point of the mixture may be greater than 500° C., such as greater than 525° C., 550° C., or 575° C. The amount of high boiling hydrocarbons, such as hydrocarbons boiling over 550° C., may be as little as 0.1 wt %, 1 wt % or 2 wt %, but can be as high as 10 wt %, 25 wt %, 50 wt % or greater. The description is explained with respect to crude oil, such as whole crude oil, but any high boiling end point hydrocarbon mixture can be used. However, processes disclosed herein can be applied to crudes, condensates and hydrocarbon with a wide boiling curve and end points higher than 500° C. Such hydrocarbon mixtures may include whole crudes, virgin crudes, hydroprocessed crudes, gas oils, vacuum gas oils, heating oils, jet fuels, diesels, kerosenes, gasolines, synthetic naphthas, raffinate reformates, Fischer-Tropsch liquids, Fischer-Tropsch gases, natural gasolines, distillates, virgin naphthas, natural gas condensates, atmospheric pipestill bottoms, vacuum pipestill streams including bottoms, wide boiling range naphtha to gas oil condensates, heavy non-virgin hydrocarbon streams from refineries, vacuum gas oils, heavy gas oils, atmospheric residuum, hydrocracker



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wax, and Fischer-Tropsch wax, among others. In some embodiments, the hydrocarbon mixture may include hydrocarbons boiling from the naphtha range or lighter to the vacuum gas oil range or heavier.

When the end boiling point of the hydrocarbon mixture is high, such as over 550° C., the hydrocarbon mixture cannot be processed directly in a steam pyrolysis reactor to produce olefins. The presence of these heavy hydrocarbons results in the formation of coke in the reactor, where the coking may occur in one or more of the convection zone preheating coils or superheating coils, in the radiant coils, or in transfer line exchangers, and such coking may occur rapidly, such as in a few hours. Whole crude is not typically cracked commercially, as it is not economical. It is generally fractionated, and only specific cuts are used in a steam pyrolysis heater to produce olefins. The remainder is used in other processes. The cracking reaction proceeds via a free radical mechanism. Hence, high ethylene yield can be achieved when it is cracked at high temperatures. Lighter feeds, like butanes and pentanes, require a high reactor temperature to obtain high olefin yields. Heavy feeds, like gas oil and vacuum gas oil (VGO), require lower temperatures. Crude contains a distribution of compounds from butanes to VGO and residue (material boiling over 550° C.). Subjecting the whole crude without separation at high temperatures produces a high yield of coke (byproduct of cracking hydrocarbons at high severity) and plugs the steam pyrolysis reactor. The steam pyrolysis reactor has to be periodically shut down and the coke is cleaned by steam/air decoking. The time between two cleaning periods when the olefins are produced is called run length. When whole crude is cracked without separation, coke can deposit in the convection section coils (vaporizing the fluid), in the radiant section (where the olefin producing reactions occur) and/or in the transfer line exchanger (where the reactions are stopped quickly by cooling to preserve the olefin yields).

Processes and systems according to embodiments herein may include a feed preparation section, a crude conditioning section, an optional aromatic complex, and a steam cracker. The feed preparation section may include a desalter, for example.

The desalted crude is then conditioned and processed such that crackable feed is being sent to the steam cracker and/or the aromatic complex. The conditioning section may allow an operator to maximize the chemicals yield while maintaining a reasonable decoking frequency of the furnaces. Another objective of the crude conditioning unit is to ensure complete or essentially complete (95%+) conversion of asphaltenes to lower boiling point components that enhance the chemicals yield while reducing the formation of heavy polynuclear aromatics (HPNAs).

Processes according to embodiments herein may thus convert heavier fractions of crude oil into high-value petrochemicals and may minimize the amount of hydrocarbons sent to a fuel oil pool, which substantially increases profitability. The small fuel oil pool that is produced may also be upgraded into a low-sulfur, IMO 2020 compliant fuel oil, further increasing the value of the products.

As noted above, high-boiling compounds in the crude oil may cause significant operational issues if they are sent to a steam cracker, due to their propensity to form coke, mainly because of their high asphaltene content. Therefore, the high boiling compounds are typically removed before sending the lighter fractions to different petrochemicals units, such as the steam cracker and aromatic complex. The removal process increases the capital cost of the overall process and lowers profitability, as the removed high-boiling compounds can

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only be sold as low-value fuel oil. In addition, conversion of vacuum residue without significant formation of HPNAs that are detrimental to steam cracker furnaces downstream of the process has been a challenge to date in the industry. Processes and systems according to embodiments herein may overcome these challenges.

The configurations of systems and processes for the conversion of whole crudes and heavy hydrocarbons according to embodiments described herein may efficiently handle resid conversion while maximizing the petrochemicals conversion and maintaining lower coking propensity in the steam cracker. This is achieved by integrating a solvent deasphalting unit to the crude conditioning unit, enabling conversion of high boiling compounds to lighter components.

The upgraded crude streams from the crude conditioning unit, such as from a fixed bed crude conditioning unit and a hydrocracker, are suitable feedstocks for the steam cracker as well as an aromatic complex, when present. Such may lead to decreasing the overall process yields of low value fuel oil and increasing the yields of high value olefins, diolefins, and aromatics, such as benzene, toluene, and xylenes (BTX).

Separation of various fractions, such as a low boiling fraction (a 160° C.- fraction, for example), a medium boiling fraction (a 160-490° C. fraction, for example), and a high boiling fraction (a 490° C.+ fraction, for example) may enhance the capital efficiency and operating costs of the processes and systems disclosed herein. While referring to three cuts in many embodiments herein, it is recognized by the present inventors that condensates, typically having a small amount of high boiling components, and whole crudes, having a greater quantity of high boiling components, may be processed differently. Accordingly, one, two, three or more individual cuts can be performed for the wide boiling range petroleum feeds, and each cut can be processed separately at optimum conditions.

Separation of the whole crude into the desired fractions may be performed using one or more separators (distillation columns, flash drums, etc.). In some embodiments, separation of the petroleum feeds may be performed in an integrated separation device (ISD), such as disclosed in US20130197283, which is incorporated herein by reference. In the ISD, an initial separation of a low boiling fraction is performed in the ISD based on a combination of centrifugal and cyclonic effects to separate the desired vapor fraction from liquid. An additional separation step may then be used to separate a middle boiling fraction from high boiling components.

Typically, hydrocarbon components boiling above 490° C. contain asphaltenes and Conradson Carbon Residue, and thus need to be processed appropriately, as described further below. While embodiments are described as including a fraction below about 90° C.-250° C., such as a 160° C.- fraction and a fraction above about 400° C.-560° C., such as a 490° C.+ fraction, it is noted that the actual cut points may be varied based on the type of whole crude or other heavy fractions being processed. For example, for a crude containing a low metals or nitrogen content, or a large quantity of "easier-to-process" components boiling, for instance, at temperatures up to 525° C., 540° C., or 565° C., it may be possible to increase the mid/high cut point while still achieving the benefits of embodiments herein. Similarly, the low/ mid cut point may be as high as 220° C. in some embodiments, or as high as 250° C. in other embodiments. Further, it has been found that a low/mid cut point of about 160° C. may provide a benefit for sizing and operation of the

reactors, such as a fixed bed conditioning reactor, for conditioning the mid fraction hydrocarbons (middle cut). Further still, for some feeds, such as condensate, the low/mid cut point may be as high as 565° C. The ability to vary the cut points may add flexibility to process schemes according to embodiments herein, allowing for processing of a wide variety of feeds while still producing the product mixture desired.

Accordingly, in some embodiments, the light cut may include hydrocarbons having a boiling point up to about 90° C. (e.g., a 90° C.- fraction), up to about 100° C., up to about 110° C., up to about 120° C., up to about 130° C., up to about 140° C., up to about 150° C., up to about 160° C., up to about 170° C., up to about 180° C., up to about 190° C., up to about 200° C., up to about 210° C., up to about 220° C., up to about 230° C., up to about 240° C., up to about 250° C. (e.g., a 250° C.- fraction), up to about 300° C., up to about 350° C., up to about 400° C., up to about 500° C., or up to about 565° C. Embodiments herein also contemplate the light cut being hydrocarbons having boiling points up to temperatures intermediate the aforementioned ranges.

Depending upon the fractionation mechanism used, the light hydrocarbon "cut" may be relatively clean, meaning the light fraction may not have any substantial amount (>1 wt % as used herein) of compounds boiling above the intended boiling temperature target. For example, a 160° C.- cut may not have any substantial amount of hydrocarbon compounds boiling above 160° C. (i.e., >1 wt %). In other embodiments, the intended target "cut" temperatures noted above may be a 95% boiling point temperature, or in other embodiments as an 85% boiling point temperature, such as may be measured using ASTM D86 or ASTM D2887, or a True Boiling Point (TBP) analysis according to ASTM D2892, for example, and ASTM D7169 for heavy streams, such as those boiling above about 400° C. In such embodiments, there may be up to 5 wt % or up to 15 wt % of compounds above the indicated "cut" point temperature. For many whole crudes, the low/mid cut point may be such that the light boiling fraction has a 95% boiling point temperature in the range from about 90° C. to about 250° C. For other feeds, however, such as condensate, the light boiling fraction may have a 95% boiling point temperature in the range from about 500° C. to about 565° C., for example.

In some embodiments, the middle cut may include hydrocarbons having a boiling point from a lower limit of the light cut upper temperature (e.g., 90° C., 100° C., 110° C., 120° C., 130° C., 140° C., 150° C., 160° C., 170° C., 180° C., 190° C., 200° C., 210° C., 220° C., 230° C., 240° C., 250° C., 300° C., 350° C., or 400° C., for example) to an upper limit of hydrocarbons having a boiling point up to about 350° C., up to about 375° C., up to about 400° C., up to about 410° C., up to about 420° C., up to about 430° C., up to about 440° C., up to about 450° C., up to about 460° C., up to about 480° C., up to about 490° C., up to about 500° C., up to about 520° C., up to about 540° C., up to about 560° C., or up to about 580° C. As used herein, for example, a middle cut having a lower limit of 160° C. and an upper limit of 490° C. may be referred to as a 160° C. to 490° C. cut or fraction. Embodiments herein also contemplate the middle cut being hydrocarbons having boiling points from and/or up to temperatures intermediate the aforementioned ranges.

Depending upon the fractionation mechanism, the hydrocarbon "cut" for the middle cut may be relatively clean, meaning the middle cut may not have any substantial amount (>1 wt %) of compounds boiling below and/or may not have any substantial amount (>1 wt %) of compounds boiling above the intended boiling temperature target limits.

For example, a 160° C. to 490° C. cut may not have any substantial amount of hydrocarbon compounds boiling below 160° C. or above 490° C. In other embodiments, the intended target "cut" temperatures noted above may be a 5 wt % or 15 wt % boiling point temperature on the lower limit and/or a 95% or 85% boiling point temperature on the upper limit, such as may be measured using ASTM D86 or ASTM D2887, or a True Boiling Point (TBP) analysis according to ASTM D2892, for example, and ASTM D7169 for heavy streams, such as those boiling above about 400° C. In such embodiments, there may be up to 5 wt % or up to 15 wt % of compounds above and/or below the "cut" point temperature, respectively.

In some embodiments, the heavy cut may include hydrocarbons having a boiling point above about 350° C., above about 375° C., above about 400° C. (e.g., a 400° C.+ fraction), above about 420° C., above about 440° C., above about 460° C., above about 480° C., above about 490° C., above about 500° C., above about 510° C., above about 520° C., above about 530° C., above about 540° C., above about 560° C., above about 580° C., above about 590° C., above about 600° C. (e.g., a 600° C.+ fraction), or above about 700° C. Embodiments herein also contemplate the heavy cut being hydrocarbons having boiling points above temperatures intermediate the aforementioned temperatures.

Depending upon the fractionation mechanism, the heavy hydrocarbon "cut" may be relatively clean, meaning the heavy fraction may not have any substantial amount (>1 wt %) of compounds boiling below the intended boiling temperature target. For example, a 490° C.+ cut may not have any substantial amount of hydrocarbon compounds boiling below 490° C. In other embodiments, the intended target "cut" temperatures noted above may be a 95% boiling point temperature, or in other embodiments as an 85% boiling point temperature, such as may be measured using ASTM D86 or ASTM D2887, or a True Boiling Point (TBP) analysis according to ASTM D2892, for example, and ASTM D7169 for heavy streams, such as those boiling above about 400° C. In such embodiments, there may be up to 5 wt % or up to 15 wt % of compounds, respectively, below the "cut" point temperature.

While examples below are given with respect to limited temperature ranges, it is envisioned that any of the temperature ranges prescribed above can be used in the processes described herein. Further, with respect to cut points, those referred to in the examples below may be clean, as described above, or may refer to 5% or 15% boiling temperatures for lower limits, or may refer to 85% or 95% boiling temperatures for upper limits.

Following fractionation, the light cut, such as a 160° C.- cut, may be fed to a steam cracker section of the system with or without further processing. The light cut fed to the steam cracker section may include light naphtha and lighter hydrocarbons, for example, and in some embodiments may include heavy naphtha boiling range hydrocarbons.

The mid-range hydrocarbon cut may be conditioned using one or more fixed bed reactors, such as hydrotreating and/or hydrocracking reactors, each of which may destructively hydrogenate the hydrocarbons in the mid-cut. The conditioning reactors may include catalysts for metals removal, sulfur removal, nitrogen removal, and the conditioning in these reactors may overall add hydrogen to the hydrocarbon components, making them easier to process downstream to produce petrochemicals. The fixed bed catalyst systems in the mid-cut conditioning zone, for example, may contain different layers of demetalizing, destructive hydrogenation and mesoporous zeolite hydrocracking catalysts to optimize

the conversion of the heavy materials to a balance between a highly paraffinic stream that is suitable for olefins production and a rich in aromatics stream that is suitable for aromatics production.

In some embodiments, it may be desirable to further separate the mid-cut into a low-mid cut and a high-mid cut. For example, a mid-cut having a boiling range from 160° C. to 490° C. may be divided into a low-mid cut having a boiling range from about 160° C. to about 325° C. and a high-mid cut having a boiling range from about 325° C. to about 490° C. The conditioning trains may thus be configured to more selectively convert the hydrocarbon components in the respective low and high mid cuts to the desired conditioned effluents, where each train may be configured based on preferred catalysts to destructively hydrogenate the hydrocarbons therein, reactor sizing for expected feed volumes and catalyst lifetime, as well as operating conditions to achieve the desired conversions to naphtha range containing steam cracker feedstocks. Similarly, division of the mid cut into three or more sub-cuts is also contemplated.

Processing of the heavy hydrocarbons, such as 490° C.+ hydrocarbons, in the solvent deasphalting unit may enhance the conversion of low value streams to high value products. The solvent deasphalting unit may further provide benefits, such as the ability to match run lengths of conditioning reactors with the steam cracker, as well as to provide an ability to handle a broader range of feeds and different crudes, allowing an operator to tune the process. The resulting solvent deasphalted oil may then be further treated, conditioning the deasphalted oil for use in the steam cracker system. It is recognized, however, that the lifetime of hydrotreating and/or hydrocracking catalysts may be negatively impacted by heavier components, such as where the feed includes components boiling above 565° C., for example.

The crude conditioning section is designed to achieve four (4) goals. First, the crude conditioning section may be used to increase the concentration of paraffins and naphthenes in the crude. Second, the conditioning section may decrease the concentration of polynuclear aromatic hydrocarbons (PNAs) in the crude. Third, the conditioning section may reduce the final boiling point (FBP) of the crude to below 540° C. And, fourth, the conditioning section may minimize the vacuum residue fraction of the crude oil.

Embodiments herein, when conditioning the middle and heavy (deasphalted oil) fractions, may target conversion of the heavier hydrocarbons to form hydrocarbons lighter than diesel, for example. The hydrotreating and hydrocracking catalysts and operating conditions may thus be selected to target the conversion of the hydrocarbons, or the hydrocarbons in the respective fractions, to primarily (>50 wt %) naphtha range hydrocarbons. In one or more embodiments, hydrotreating and hydrocracking catalysts and operating conditions may thus be selected to target the conversion of the hydrocarbons, or the hydrocarbons in the in the respective fractions, to primarily (>50 wt %) steam crackable products. The use of catalysts and operating conditions in the conditioning section to target lighter hydrocarbon products may enhance the operability of the steam cracker and the production of chemicals.

In some embodiments, conditioning of the heavy cut, such as a 490° C.+ cut, may result in conversion of at least 70 wt % of the compounds boiling in excess of 565° C. to lighter boiling compounds. Other embodiments may result in conversion of greater than 75 wt %, greater than 80 wt %, or greater than 85 wt % of the compounds boiling in excess of 565° C. to lighter boiling compounds.

In some embodiments, conditioning of the middle cut, such as a 160° C. to 490° C. cut, may result in conversion of greater than 50 wt % of the hydrocarbons therein to naphtha range hydrocarbons. In other embodiments, conditioning of the middle cut may result in conversion of greater than 55 wt %, greater than 60 wt %, or greater than 65 wt %, or greater than 70 wt % of the hydrocarbons therein to naphtha range hydrocarbons.

In some embodiments, collective conditioning of the middle cut and the heavy cut may result in an overall conversion of greater than 50 wt % of the hydrocarbons therein to naphtha range hydrocarbons. In other embodiments, conditioning of the middle cut and the heavy cut may result in conversion of greater than 55 wt %, greater than 60 wt %, or greater than 65 wt % of the hydrocarbons therein to naphtha range hydrocarbons.

As a result of such initial separations and conditioning, feeds to the steam cracker may be fed, in some embodiments, directly to the steam cracker without further processing. The light cut, having preferred properties, including one or more of boiling point, API, BMCI, hydrogen content, nitrogen content, sulfur content, viscosity, MCRT, or total metals content, may be fed directly to the steam cracker following separations in some embodiments. Effluents from the middle cut conditioning may also be fed directly to the steam cracker according to embodiments herein. Likewise, effluents from the heavy cut conditioning may be fed directly to the steam cracker in some embodiments.

The conditioning of the respective fractions as described herein may allow for the steam cracker, even while processing multiple feeds of varying boiling point ranges, to run for an extended period of time. In some embodiments, the steam cracker may be able to run for an uninterrupted run length of at least three years; at least four years in other embodiments; and at least five years in yet other embodiments.

Further, the initial hydrocarbon cut points, reactor sizes, catalysts, etc. may be adjusted or configured such that a run time of the steam cracker operations and conditioning processes may be aligned. The catalysts, reactor sizes, and conditions may be configured such that a run time of the conditioning unit is aligned with the run time of the steam cracker. Catalyst volumes, catalyst types, and reaction severity may all play a role in determining conditioning unit run times. Further, the extent of conditioning of the heavier hydrocarbons in the crude may impact coking in the thermal cracker. To maximize plant uptime, embodiments herein contemplate design and configuration of the overall system such that the conditioning system has a similar anticipated run time as the steam cracker for a given feedstock or a variety of anticipated feedstocks. Further, embodiments herein contemplate adjustment of reaction conditions (T, P, space velocity, etc.) in the conditioning section and/or the steam cracker based on a feedstock being processed, such that a run time of the conditioning section and the steam cracker are similar or aligned.

Alignment of run times may result in minimal downtime, such as where a catalyst turnover in a conditioning reactor is conducted concurrently with decoking of the steam cracker. Where the conditioning system includes multiple reactors or types of reactors, alignment of the run times may be based on the expected steam cracker performance. Further, where a hydrotreater, for example, may have a significantly longer run time than a hydrocracker in the conditioning section, parallel reactor trains and/or bypass processing may be used such that the overall run times of the conditioning and steam cracking units may be aligned.

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Bypass processing may include, for example, temporarily processing a 490° C.+ cut in a reactor which normally processes a lighter feedstock. The heavier feedstock is anticipated to have more severe conditions and shorter catalyst life, and thus temporarily processing the heavies in a mid-range hydrocarbons conditioning reactor during a heavies catalyst change may allow the whole crude feed to continue to be fed to the steam cracker, without a shutdown, while the heavies conditioning reactor catalyst is replaced. Configuration of the mid-range conditioning reactors may also take into account the anticipated bypass processing when designing the overall system for aligned run times.

Referring now to FIG. 1, a simplified process flow diagram of a system for converting whole crudes and heavy hydrocarbons according to embodiments herein is illustrated.

A wide boiling range heavy hydrocarbon feed, such as a desalted crude 1, may be fed to a separation system 3. Separation system 3 may be an integrated separation device (ISD), as described above and including separation and heat integration, for example. In separation system 3, the desalted crude 1 may be separated into three fractions, including (a) a 160° C.- fraction 5 that doesn't require any conditioning and can be used as feed to the steam cracker section 7, (b) a 160-490° C. fraction 9 that may be upgraded in a conditioning section 11 to produce lighter hydrocarbons, such as a highly paraffinic stream 13 suitable for processing in the steam cracking section 7 and, (c) a 490° C.+ fraction 15, which contains the most refractory materials in the crude, and which can be upgraded in a solvent deasphalting system 17 in association with hydrotreaters and/or hydrocrackers 27. Other cut points may also be used to route the desired fractions and hydrocarbons therein to desired units for conditioning and/or cracking. The solvent deasphalting system 17 may produce a pitch 19 and a desphalted oil 21 that is suitable to be further processed in the conditioning system to produce additional hydrocarbons suitable for conversion to chemicals in the steam cracker section 7. The processing of the feeds in the steam cracker section may produce one or more chemical streams 23, such as ethylene, propylene, and butenes, among others, as well as a higher boiling pyrolysis oil fraction 25.

In some embodiments, the mid cut (160-490° C.) stream may be processed initially in a fixed bed conditioning system 27. The 490° C.+ stream may be processed in the solvent deasphalting system 17, which may include one or more solvent deasphalting units, to produce a desphalted oil. The desphalted oil may be further processed in the conditioning section 11, which may include the same reactor (s) used to condition the mid cut, as illustrated, or a separate fixed bed conditioning system 29 (FIG. 2) that may contain a catalyst tailored to effectively condition the desphalted oil received from the solvent deasphalting system. The reaction products 13 from the fixed bed conditioning reactor 27 may then be co-processed in steam cracker section 7 for conversion into light olefins and other valuable chemicals. The pitch may be processed in one or more downstream processes (not illustrated).

Referring now to FIG. 2, a simplified process flow diagram of a system for converting whole crudes and heavy hydrocarbons according to embodiments herein is illustrated, where like numerals represent like parts. In this embodiment, the desalted whole crude is processed similar to that as described above for FIG. 1. In this embodiment, the 490° C.+ stream 15 is processed in a solvent deasphalting system 17, and the desphalted oil 21 is processed in a second fixed bed conditioning system 29. The separate

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conditioning of the mid and deasphalted oil fractions may permit an operator to better tailor the catalyst to the respective feeds, as well as to size the reactors appropriately.

As described above, the fixed bed conditioning systems 27, 29 may be used to condition the 160-490° C. mid cut stream 9 and the deasphalted oil stream 21 separately. In some embodiments, the streams may be processed in the same hydrotreating reactor (as in FIG. 1). However, it has been found that, due to the nature of the feed compounds for various crudes, processing in a single reaction train may result in a stream with molecules that contain more aromatic rings than the molecules in straight run Arab Light or Arab Extra Light crudes in the same boiling range. As a result, more severe hydrotreating conditions may be necessary to sufficiently saturate the molecules, which has a negative impact on hydrotreating catalyst life and/or capital investment. If the deasphalted oil in stream 21 are co-processed with the straight run 160-490° C. material in stream 9, the turnaround time for a single hydrotreating train may drop undesirably, and/or a spare hydrotreating train would be required to provide a steady stream of feed to the steam cracking section while the hydrotreating catalyst system is undergoing regeneration and/or replacement. The aforementioned would also be applicable to other types of crude, such as desalted oils, condensate, biogenic oil, synthesis crude, tight oil, heavy hydrocarbons, reconstituted crudes, and bitumen derived oils.

To alleviate the issues of catalyst life/turnaround time, the fixed bed hydrotreating step may be split into separate trains, as illustrated in FIG. 2. One train may be provided for processing the straight run 160-490° C. crude oil, and a second train may be provided for processing the deasphalted oil. Generally, the reactors 27 in the first hydrotreating system may have less frequent turnaround time than that of reactors 29 in the second train, which may have more frequent turnarounds to replace catalyst, but the straight run 160-490° C. and deasphalted oil could be combined to either the first train (27) or the second train (29), during catalyst replacement, so as to not require a spare reactor train for uptime. As a temporary diversion of feed, the impact on either reactor train would be minimal, and thus reactor trains 27 could be designed such that its turnarounds may be in sync with that of the steam cracker furnaces.

As noted above, various feedstocks may allow the cut points to be increased, such as raising the mid/high cut point from 490° C. to 545° C. in some embodiments. The same may be true with respect to processing in the solvent deasphalting system, where higher boiling point hydrocarbons may be recovered with the deasphalted oil and fed to the hydrotreating reactor for conversion into feedstocks suitable for steam cracking. However, with respect to processing of the high boiling fraction (e.g., 490° C.+ or 545° C.+ fraction) in the solvent deasphalting system, it has been found that a lower cut point may be more favorable.

Referring now to FIG. 3, a simplified process flow diagram of a system for converting whole crudes and heavy hydrocarbons according to embodiments herein is illustrated, where like numerals represent like parts. In this embodiment, the desalted whole crude is processed similar to that as described above for FIG. 2. In this embodiment, the pyrolysis oil stream 25 may be processed in the solvent deasphalting system 17. The pyrolysis oil will be removed with the desphalted oil 21, increasing the quantity of upgradable hydrocarbons being fed to fixed bed conditioning system 29, and ultimately to steam cracking system 7, which may further increase the conversion of crude feedstock 1 to one or more chemical products 23. Recycling the

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pyrolysis oil stream **25** to the solvent deasphalting unit **17** may also prevent the fixed bed conditioning system **29** from running dry in embodiments where the 490° C.+ or 545° C.+ fraction **15** is substantially pitch.

As described briefly above, embodiments herein may allow for the direct cracking of crude oil to chemicals, forming light hydrocarbons like ethylene, propylene and light aromatics, in an economically viable manner, without passing through the conventional refining steps. Additionally, direct conversion of crude oil to chemicals may help close the widening supply-demand gap for key building blocks normally produced as co-products (propylene, butadiene) due to the increasing shift toward cracking lighter feedstock spurred by the shale gas revolution.

Integration of processing units according to embodiments herein may provide the unique potential for upgrading whole crudes, such as Arab Light crude and Arab Extra Light crude, along with low value refinery streams, such as Pyrolysis Oil (PyOil), slurry oil and Light Cycle Oil (LCO), into higher value chemical products. While conditioning of the feeds according to embodiments herein adds hydrogen to the feed components, and the hydrogen consumption is an added expense to the plant, the overall benefits in producing chemicals, rather than fuels, outweighs this added expense. The aforementioned would also be applicable to other types of crude, such as desalted oils, condensate, biogenic oil, synthesis crude, tight oil, heavy hydrocarbons, reconstituted crudes, and bitumen derived oils.

In other embodiments, an optional aromatics complex may be included. For example, an aromatics complex may be used to convert the 160° C.-490° C. fraction, or a portion thereof, to aromatics. For example, a mid-cut such as 160° C. to 240° C. fraction may be processed to convert a portion of the hydrocarbons therein to aromatics, while the heavies may be fed to the steam cracker for conversion to chemicals. The aromatics complex feedstock generated via initial processing and conditioning according to embodiments herein may permit various processors to discontinue importing full range naphtha (FRN).

Further, in some embodiments, the pyrolysis oil generated in the steam cracking unit may be separated to recover a pyrolysis gasoline fraction, and one or more heavies fractions, such as a pyrolysis gas oil fraction and a pyrolysis fuel oil fraction. The lighter pyrolysis gasoline fraction may be fed to an aromatics unit, while the heavier fractions may be used to form an ULSCO, as noted above.

As described with respect to FIGS. 1-3, separation system **3** may be a separation system as illustrated in FIG. 4, including separation and heat integration. After desalting, the crude **1** may be further preheated in the convection section of a heater **500** to produce a preheated crude **502**. The preheated crude **502** may be fed to a separator **504** which may facilitate the separation of the 160° C.- fraction **5**.

The remaining 160° C.+ crude fraction **506** may be fed to a pump **508**, which produces a pressurized 160° C.+ crude fraction **510**, which may then be fed to a heat exchanger **512**. Heat exchanger **512** may preheat the 160° C.+ crude fraction **510** against hot hydrogen stripper bottoms **520**, producing a pressurized and pre-heated 160° C.+ crude fraction **514**. The pressurized and pre-heated 160° C.+ crude fraction **514** may then be fed back to the heater **500** where it is heated to facilitate the separation of a 160-490° C. fraction from a heavier 490° C.+ fraction. The heated 160° C.+ crude fraction **516** may then be fed to a hot hydrogen stripper **518**. In the hot hydrogen stripper **518**, the 160° C.+ crude fraction is further separated into a 160-490° C. fraction **9** and the hot hydrogen stripper bottoms **520**, which contains heavier 490°

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C.+ hydrocarbons. The hot hydrogen stripper bottoms **520**, after being cooled via indirect heat exchange in heat exchanger **512** against the pressurized 160° C.+ crude fraction **510**, may be removed from the separation system **3** as the 490° C.+ fraction **15**.

The hot hydrogen stripper **518** may utilize a hydrogen feed **522** as the stripping medium. The hot hydrogen stripper **518** may be operated to provide broad flexibility, based on the nature of the crude feedstock that is being processed. The stripper overheads, which is the 160-490° C. fraction **9**, may be cooled, to recover hydrogen, and routed to the intermediate hydroprocessing reaction stages as appropriate, and as described with respect with FIGS. 1-3. The recovered hydrogen may be fed to a downstream pressure swing adsorption (PSA) unit (not shown), after amine treatment (not shown), to improve the hydrogen purity. The PSA hydrogen product may be compressed in a make-up hydrogen compressor (not shown) to provide the make-up hydrogen for the one or more hydroprocessing reactors (FIGS. 1-3), and as hot hydrogen feed **522**.

The hot hydrogen stripper bottoms product **520** (such as a 490° C.+ cut) contains the most difficult compounds which must be handled in the crude, including asphaltenes, metals, and CCR. The excessive amount of metals, CCR, and asphaltenes in the high boiling residue fraction leads to rapid fouling of catalyst and increase of pressure drop in fixed bed down-flow reactors, limiting both conversion and catalyst run length. After cooling against the pressurized 160° C.+ crude fraction **510**, the 490° C.+ stream **15** may be recovered and processed in a liquid circulation, ebullated bed residue hydrocracker, as described in FIGS. 1-3, along with any additional low value refinery streams, such as a pyoil stream and/or slurry oil stream.

By adjusting the amount of hydrogen **522** fed to the hot hydrogen stripper **518**, as well as the operating conditions of the hot hydrogen stripper **518** and heater **500**, the hydrocarbon cut points may be adjusted such that the light-cut **5** may be fed directly to the downstream steam cracker, and the mid-cut **9** may have little to no deleterious compounds that would rapidly foul the fixed bed conditioning reactors. In this way, the separation system **3** (with the hot hydrogen stripper **518**) may concentrate the most difficult to process hydrocarbons in the heavy-cut **15** which may be fed to the ebullated bed reactors which may be operated at the most severe conditions, thereby preserving the catalysts in the steam cracker and fix bed conditioning reactors. Embodiments herein provide a strategic combination of crude feed preparation, crude separation, crude conditioning, and steam cracking technology to maximize the yield of high value chemicals. The crude conditioning section employs a combination of fixed bed hydroprocessing and liquid circulation to condition the crude into a suitable steam cracker feed and to upgrade the low value refinery streams. Embodiments herein may achieve a yield of chemicals in the range of 60% to 90% of the whole crude feedstock, for example.

After desalting, the crude may be segregated into three cuts, including: a 160° C.- stream; a 160-490° C. stream; and a 490° C.+ stream. The 160° C.- stream does not require upgrading, and thus can be directly routed as steam cracker. The 160-490° C. stream is easily handled in a fixed bed hydroprocessing reaction system, in which the feed is hydrotreated and converted to naphtha, making an ideal steam cracker feedstock.

Embodiments herein may employ a solvent deasphalting unit to remove the pitch (asphaltene) and metals, thereby increasing the runtime of the conversion process without fouling the reactors, while the pitch, asphaltene, and metals

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may be fed to a delayed coking unit to recover the carbon that is contained in these streams.

Embodiments herein may provide upstream processing to process whole crude and other wide boiling range hydrocarbons in a steam cracker, where embodiments of the overall integrated processes may be configured to have a common run time. Avoiding entrainment of heavy materials in the front end separations may lower costs, and may result in less complex flow schemes as illustrated and described herein. Further, the avoidance of entrainment may ensure the operability and processability in the crude conditioning systems and steam cracker, allowing for lower overall capital expense while achieving a high yield of chemicals. This may be done by having fail-over, or cut-over, from one conditioning system to the other in order to minimize total system downtime during catalyst regeneration, maintenance, or cleaning. Further, such embodiments may eliminate the need for parallel reaction trains or redundant process units, in both the mid-range hydrocarbon processing and the high boiling residue processing, for use during catalyst regeneration.

Further, the hydrotreating and hydrocracking reactors in each of the conditioning systems may be sized to have run-times similar to the steam cracking unit. Such configurations may additionally allow for reduced down-time as cleaning, maintenance, and catalyst regeneration may be accomplished all at once across multiple reaction systems. Without such design considerations, operations may have increased downtime while reactors in the conditioning system, for example, are taken offline for catalyst regeneration while the catalysts in the second conditioning system still have >50% catalyst life.

Additionally, avoiding entrainment of heavy materials in the front end separations may lower costs, and may result in less complex flow schemes as illustrated and described herein. Further, the avoidance of entrainment may ensure the operability and processability in the crude conditioning systems and steam cracker, allowing for lower overall capital expense while achieving a high yield of chemicals.

As described above, embodiments herein may separate a desalted crude or other wide boiling hydrocarbons into various fractions to effectively condition the respective fractions to form a feedstock suitable for conversion in a steam cracker. Because of the wide range of feedstocks that may be processed according to embodiments herein, depending upon the feedstock, conditioning catalysts, reactor volumes, and other factors for a given installation, it may be more preferential to base the specific cut points based on one or more additional properties of the feedstock. For example, the specific cut points may be adjusted based on one or more properties or additional properties of the crude feedstock, such as API gravity, Bureau of Mines Correlation Index (BMCI), hydrogen content, nitrogen content, sulfur content, viscosity, microcarbon residue (MCRT), and/or total metals, among other feedstock properties.

Various feedstocks useful in embodiments herein, such as crude oils, desalted oils, condensate, biogenic oil, synthetic crude, tight oil, heavy hydrocarbons, reconstituted crude and bitumen derived oil may have one or more of the following properties, including: an API gravity between 4 and 60°, a BMCI of 20 to 85, a hydrogen content of 9.0 to 14.5 wt % (or 90,000 to 145,000 ppm), a nitrogen content of 0.02 to 0.95 wt % (or 200 to 9,500 ppm), a sulfur content of 0.009 to 6.0 wt % (or 90 to 60,000 ppm), a viscosity, at 40° C., of 95 to 5500 centistokes (cSt), a MCRT of 5 to 35 wt %, and/or may have a total metals content of <1 to 1000 ppm.

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The initial crude separations may be conducted and adjusted in order to have the light-, mid-, and heavy-cuts have specific properties, such that the light-cut may go to the steam cracker with no, or minimal, intermediate processing. Further, the mid to heavy cuts may be conducted and adjusted in order to have the mid-cut and heavy-cut have appropriate and/or favorable feed properties and hydrocarbon species so as to be effectively and efficiently conditioned in the mid and heavy conditioning reactors.

#### BMCI

In some embodiments, the light cut may have a BMCI of less than 20. In other embodiments, the light cut may have a BMCI of less than 15. In yet other embodiments, the light cut may have a BMCI of less than 10 or even less than 5. In some embodiments, the mid cut may have a BMCI of less than 40, such as less than 35, less than 30, or less than 25. In some embodiments, the heavy cut may have a BMCI of greater than 30, such as greater than 35, greater than 40, greater than 45, greater than 50, or greater than 55.

Accordingly, in some embodiments, a light cut, including hydrocarbons having a boiling point up to about 90° C. to about 300° C., for example, may have a BMCI of less than 20; in other embodiments, such as when the light cut includes hydrocarbons having a boiling point up to about 110° C. or up to about 250° C., for example, the light cut may have a BMCI of less than 10; in yet other embodiments, such as when the light cut includes hydrocarbons having a boiling point up to about 130° C. or up to about 220° C., for example, the light cut may have a BMCI of less than 5. In some embodiments where the light cut includes hydrocarbons having a boiling point below about 160° C., the light cut may have a BMCI of less than 5. While the BMCI may vary for the different feeds at any given cut temperature, a low BMCI, such as less than 10 or less than 5, for example, has been found to improve the processability of the light hydrocarbons in the steam pyrolysis unit without the need for intermediate processing. Light cuts for Arab light crudes processed according to embodiments herein may target a BMCI of less than 10, for example, and may target a BMCI of less than 6 or less than 5.5 for Arab extra light crudes, for example.

In some embodiments, the mid cut, including hydrocarbons having a lower boiling point in the range from about 90° C. to about 300° C. and an upper boiling point in the range from about 400° C. to about 600° C., may have a BMCI of between about 5 and 50. For example, the mid cut may have a BMCI of between a lower limit of 5, 10, 15, 20, or 25 to an upper limit of 10, 15, 20, 25, 30, 40, or 50. A mid-cut having a BMCI of between 10 and 30, for example, has been found to be convertible to steam cracker feeds using relatively moderate destructive hydrogenation conditions in the mid-cut conditioning section of processes herein. Mid-cuts for Arab light crudes processed according to embodiments herein may target a BMCI in the range from about 20 to about 30, for example, and may target a BMCI in the range from about 15 to about 30 for Arab extra light crudes, for example.

In various embodiments, the heavy cut, including hydrocarbons having a boiling point greater than about 300° C., may have a BMCI of greater than 30. When the heavy cut includes hydrocarbons having a boiling point above about 350° C., the heavy cut may have a BMCI of greater than 40. When the heavy cut includes hydrocarbons having a boiling point above about 400° C., the heavy cut may have a BMCI of greater than 50. In embodiments where the heavy cut includes hydrocarbons having a boiling point above about 490° C., the heavy cut may have a BMCI of greater than 55.

A heavy-cut having a BMCI of greater than about 40, for example, has been found to be convertible to steam cracker feeds using the more severe destructive hydrogenation conditions in the heavy-cut conditioning section of processes herein. Heavy-cuts for Arab light crudes processed according to embodiments herein may target a BMCI in the range from about 50 to about 60, for example, and may target a BMCI in the range from about 25 to about 40 for Arab extra light crudes, for example.

#### API

In some embodiments, the light cut may have an API gravity of greater than 10°. In other embodiments, the light cut may have an API gravity of greater than 15°. In yet other embodiments, the light cut may have an API gravity of greater than 20°, greater than 30°, or even greater than 40°. In some embodiments, the mid cut may have an API gravity of greater than 10° and less than 40°, such as from a lower limit of 10°, 15°, 20°, 25°, or 30° to an upper limit of 25°, 30°, 35°, 40°, 45°, or 50°. In some embodiments, the heavy cut may have an API gravity of less than 40°, such as less than 35°, less than 25°, less than 20°, less than 15°, or less than 10°.

Accordingly, in some embodiments, a light cut, including hydrocarbons having a boiling point up to about 300° C., for example, may have an API gravity of greater than 10°; in other embodiments, such as when the light cut includes hydrocarbons having a boiling point up to about 250° C., for example, the light cut may have an API gravity of greater than 20°; in yet other embodiments, such as when the light cut includes hydrocarbons having a boiling point up to about 220° C., for example, the light cut may have an API gravity of greater than 40°. In some embodiments where the light cut includes hydrocarbons having a boiling point below about 160° C., the light cut may have an API gravity of greater than 60°. While the API gravity may vary for the different feeds at any given cut temperature, an API gravity, such as greater than 40°, greater than 50°, or greater than 60°, for example, has been found to improve the processability of the light hydrocarbons in the steam pyrolysis unit without the need for intermediate processing. Light cuts for Arab light crudes processed according to embodiments herein may target an API gravity of greater than 65°, for example, and may target an API gravity of greater than 60° for Arab extra light crudes, for example.

In some embodiments, the mid cut, including hydrocarbons having a lower boiling point in the range from about 90° C. to about 300° C. and an upper boiling point in the range from about 400° C. to about 600° C., may have an API gravity of between about 5° and 50°. For example, the mid cut may have an API gravity of between a lower limit of 5°, 10°, 15°, 20°, or 25° to an upper limit of 10°, 15°, 20°, 25°, 30°, 40°, or 50°. A mid-cut having an API gravity of between 20° and 40°, for example, has been found to be convertible to steam cracker feeds using relatively moderate destructive hydrogenation conditions in the mid-cut conditioning section of processes herein. Mid-cuts for Arab light crudes processed according to embodiments herein may target an API gravity in the range from about 30° to about 35°, for example, and may target an API gravity in the range from about 35° to about 40° for Arab extra light crudes, for example.

In various embodiments, the heavy cut, including hydrocarbons having a boiling point greater than about 300° C., may have an API gravity of less than about 40°. When the heavy cut includes hydrocarbons having a boiling point above about 350° C., the heavy cut may have an API gravity of less than about 20°. When the heavy cut includes hydro-

carbons having a boiling point above about 400° C., the heavy cut may have an API gravity of less than about 10°. In embodiments where the heavy cut includes hydrocarbons having a boiling point above about 490° C., the heavy cut may have an API gravity of less than 7°, for example. A heavy-cut having an API gravity of less than about 20°, for example, has been found to be convertible to steam cracker feeds using the more severe destructive hydrogenation conditions in the heavy-cut conditioning section of processes herein. Heavy-cuts for Arab light crudes processed according to embodiments herein may target an API gravity in the range from about 5° to about 10°, for example, and may target an API gravity in the range from about 10° to about 20° for Arab extra light crudes, for example.

#### Hydrogen Content

In some embodiments, the light cut may have a hydrogen content of greater than 12 wt %. In other embodiments, the light cut may have a hydrogen content of greater than 13 wt %. In yet other embodiments, the light cut may have a hydrogen content of greater than 13.5 wt %, greater than 14 wt %, or even greater than 15 wt %. In some embodiments, the mid cut may have a hydrogen content of greater than 11 wt % and less than 14 wt %, such as from a lower limit of 11, 11.5, 12.0, 12.5, or 13.0 wt % to an upper limit of 12.0, 12.5, 13.0, 13.5, 14.0, or 14.5 wt %. In some embodiments, the heavy cut may have a hydrogen content of less than 13 wt %, such as less than 12.5 wt %, less than 12 wt %, less than 11.5 wt %, or less than 11 wt %.

Accordingly, in some embodiments, a light cut, including hydrocarbons having a boiling point up to about 300° C., for example, may have an hydrogen content of greater than 13 wt %; in other embodiments, such as when the light cut includes hydrocarbons having a boiling point up to about 250° C., for example, the light cut may have an hydrogen content of greater than 13.5 wt %; in yet other embodiments, such as when the light cut includes hydrocarbons having a boiling point up to about 220° C., for example, the light cut may have an hydrogen content of greater than 14.0 wt %. In some embodiments where the light cut includes hydrocarbons having a boiling point below about 160° C., the light cut may have a hydrogen content of greater than 14.5 wt %. While the hydrogen content may vary for the different feeds at any given cut temperature, a hydrogen content, such as greater than 13 wt %, greater than 14 wt %, or greater than 14.5 wt %, for example, has been found to improve the processability of the light hydrocarbons in the steam pyrolysis unit without the need for intermediate processing. Light cuts for Arab light crudes processed according to embodiments herein may target a hydrogen content of greater than 14.5 wt %, for example, and may target a hydrogen content of greater than 14 wt % for Arab extra light crudes, for example.

In some embodiments, the mid cut, including hydrocarbons having a lower boiling point in the range from about 90° C. to about 300° C. and an upper boiling point in the range from about 400° C. to about 600° C., may have a hydrogen content of between about 11.5 wt % and 14.5 wt %. A mid-cut having a hydrogen content of between 12 wt % and 13.5 wt %, for example, has been found to be convertible to steam cracker feeds using relatively moderate destructive hydrogenation conditions in the mid-cut conditioning section of processes herein. Mid-cuts for Arab light crudes processed according to embodiments herein may target a hydrogen content in the range from about 12.5 wt % to about 13.5 wt %, for example, and may target an hydrogen content in the range from about 13.0 wt % to about 14.0 wt % for Arab extra light crudes, for example.



In various embodiments, the heavy cut, including hydrocarbons having a boiling point greater than about 300° C., may have a hydrogen content of less than about 13 wt %. When the heavy cut includes hydrocarbons having a boiling point above about 350° C., the heavy cut may have a hydrogen content of less than about 12.5 wt %. When the heavy cut includes hydrocarbons having a boiling point above about 400° C., the heavy cut may have a hydrogen content of less than about 12.0 wt %. In embodiments where the heavy cut includes hydrocarbons having a boiling point above about 490° C., the heavy cut may have a hydrogen content of less than 11 wt %, for example. A heavy-cut having a hydrogen content of less than about 12 wt %, for example, has been found to be convertible to steam cracker feeds using the more severe destructive hydrogenation conditions in the heavy-cut conditioning section of processes herein. Heavy-cuts for Arab light crudes processed according to embodiments herein may target a hydrogen content in the range from about 10 wt % to about 11 wt %, for example, and may target a hydrogen content in the range from about 11 wt % to about 12 wt % for Arab extra light crudes, for example.

#### Nitrogen Content

In some embodiments, the light cut may have a nitrogen content of less than 100 ppm, such as less than 50 ppm or less than 30 ppm. In other embodiments, the light cut may have a nitrogen content of less than 25 ppm. In yet other embodiments, the light cut may have a nitrogen content of less than 20 ppm, less than 15 ppm, less than 10 ppm, less than 5 ppm, less than 3 ppm, less than 1 ppm, or even less than 0.5 ppm. In some embodiments, the mid cut may have a nitrogen content of greater than 1 ppm and less than 1000 ppm, such as from a lower limit of 1, 5, 10, 50, 100, 250, or 500 ppm to an upper limit of 50, 100, 250, 500, or 1000 ppm. In some embodiments, the heavy cut may have a nitrogen content of greater than 10 ppm, such as greater than 25 ppm, greater than 50 ppm, greater than 100 ppm, greater than 150 ppm, greater than 200 ppm, greater than 250 ppm, greater than 500 ppm, greater than 1000 ppm, greater than 1500 ppm, greater than 2000 ppm, or greater than 2500 ppm.

Accordingly, in some embodiments, a light cut, including hydrocarbons having a boiling point up to about 300° C., for example, may have an nitrogen content of less than 0.01 wt %, or 100 ppm; in other embodiments, such as when the light cut includes hydrocarbons having a boiling point up to about 250° C., for example, the light cut may have an nitrogen content of less than 0.001 wt %, or 10 ppm; in yet other embodiments, such as when the light cut includes hydrocarbons having a boiling point up to about 220° C., for example, the light cut may have a nitrogen content of less than 0.0001 wt %, or 1 ppm. In some embodiments where the light cut includes hydrocarbons having a boiling point below about 160° C., the light cut may have a nitrogen content of less than about 0.00003 wt %, or 0.3 ppm. While the nitrogen content may vary for the different feeds at any given cut temperature, a nitrogen content, such as less than about 100 ppm, less than 10 ppm, or less than 1 ppm, for example, has been found to improve the convertibility of the light hydrocarbons in the steam pyrolysis unit without the need for intermediate processing. Light cuts for Arab light crudes processed according to embodiments herein may target a nitrogen content of less than 1 ppm, for example, and may also target a nitrogen content of less than 1 ppm for Arab extra light crudes, for example.

In some embodiments, the mid cut, including hydrocarbons having a lower boiling point in the range from about 90° C. to about 300° C. and an upper boiling point in the

range from about 400° C. to about 600° C., may have a nitrogen content of between about 10 ppm and 250 ppm, for example. A mid-cut having a nitrogen content of between 20 and 250 ppm, for example, has been found to be convertible to steam cracker feeds using relatively moderate destructive hydrogenation conditions in the mid-cut conditioning section of processes herein. Mid-cuts for Arab light crudes processed according to embodiments herein may target a nitrogen content in the range from about 200 to about 300 ppm, for example, and may target an nitrogen content in the range from about 100 to about 150 ppm for Arab extra light crudes, for example.

In various embodiments, the heavy cut, including hydrocarbons having a boiling point greater than about 300° C., may have a nitrogen content of greater than about 0.001 wt %, or 10 ppm. When the heavy cut includes hydrocarbons having a boiling point above about 350° C., the heavy cut may have a nitrogen content of greater than about 0.005 wt %, or 50 ppm. When the heavy cut includes hydrocarbons having a boiling point above about 400° C., the heavy cut may have a nitrogen content of greater than about 0.01 wt %, or 100 ppm. In embodiments where the heavy cut includes hydrocarbons having a boiling point above about 490° C., the heavy cut may have a nitrogen content of greater than 2500 ppm, for example. A heavy-cut having a nitrogen content of greater than about 100 ppm, for example, has been found to be convertible to steam cracker feeds using the more severe destructive hydrogenation conditions in the heavy-cut conditioning section of processes herein. Heavy-cuts for Arab light crudes processed according to embodiments herein may target a nitrogen content in the range from about 2000 to about 3000 ppm, for example, and may target a nitrogen content in the range from about 1000 to about 2000 for Arab extra light crudes, for example.

#### Sulfur Content

In some embodiments, the light cut may have a sulfur content of less than 10000 ppm, such as less than 5000 ppm or less than 1000 ppm. In other embodiments, the light cut may have a sulfur content of less than 750 ppm. In yet other embodiments, the light cut may have a sulfur content of less than 500 ppm, less than 250 ppm, or even less than 100 ppm. In some embodiments, the mid cut may have a sulfur content of greater than 500 ppm and less than 10000 ppm, such as from a lower limit of 500, 750, 1000, 1500, 2000, 2500, or 5000 ppm to an upper limit of 1000, 2000, 5000, 10000, 15000, or 20000 ppm. In some embodiments, the heavy cut may have a sulfur content of greater than 1000 ppm, such as greater than 2500 ppm, greater than 5000 ppm, greater than 10000 ppm, greater than 15000 ppm, greater than 20000 ppm, greater than 25000 ppm, greater than 30000 ppm, greater than 35000 ppm, greater than 40000 ppm, greater than 45000 ppm, or greater than 50000 ppm.

Accordingly, in some embodiments, a light cut, including hydrocarbons having a boiling point up to about 300° C., for example, may have an sulfur content of 1 wt %, or 10,000 ppm; in other embodiments, such as when the light cut includes hydrocarbons having a boiling point up to about 250° C., for example, the light cut may have an sulfur content of less than 0.5 wt %, or 5,000 ppm; in yet other embodiments, such as when the light cut includes hydrocarbons having a boiling point up to about 220° C., for example, the light cut may have a sulfur content of less than 0.1 wt %, or 1,000 ppm. In some embodiments where the light cut includes hydrocarbons having a boiling point below about 160° C., the light cut may have a sulfur content of less than about 750 ppm or less than 500 ppm. While the sulfur content may vary for the different feeds at any given cut



temperature, a sulfur content, such as less than about 600 ppm, for example, has been found to improve the convertibility of the light hydrocarbons in the steam pyrolysis unit without the need for intermediate processing. Light cuts for Arab light crudes processed according to embodiments herein may target a sulfur content of less than 750 ppm, for example, and may also target a sulfur content of less than 500 ppm for Arab extra light crudes, for example.

In some embodiments, the mid cut, including hydrocarbons having a lower boiling point in the range from about 90° C. to about 300° C. and an upper boiling point in the range from about 400° C. to about 600° C., may have a sulfur content of between about 1000 ppm and 20000 ppm, for example. A mid-cut having a sulfur content of between 2000 and 15000 ppm, for example, has been found to be convertible to steam cracker feeds using relatively moderate destructive hydrogenation conditions in the mid-cut conditioning section of processes herein. Mid-cuts for Arab light crudes processed according to embodiments herein may target a sulfur content in the range from about 6000 to about 12000 ppm, for example, and may target an sulfur content in the range from about 5000 to about 10000 ppm for Arab extra light crudes, for example.

In various embodiments, the heavy cut, including hydrocarbons having a boiling point greater than about 300° C., may have a sulfur content of greater than about 0.1 wt %, or 1,000 ppm. When the heavy cut includes hydrocarbons having a boiling point above about 350° C., the heavy cut may have a sulfur content of greater than about 0.5 wt %, or 5,000 ppm. When the heavy cut includes hydrocarbons having a boiling point above about 400° C., the heavy cut may have a sulfur content of greater than about 1 wt %, or 1,0000 ppm. In embodiments where the heavy cut includes hydrocarbons having a boiling point above about 490° C., the heavy cut may have a sulfur content of greater than 25000 ppm, for example. A heavy-cut having a sulfur content of greater than about 10000 ppm, for example, has been found to be convertible to steam cracker feeds using the more severe destructive hydrogenation conditions in the heavy-cut conditioning section of processes herein. Heavy-cuts for Arab light crudes processed according to embodiments herein may target a sulfur content in the range from about 30000 to about 50000 ppm, for example, and may target a sulfur content in the range from about 20000 to about 30000 for Arab extra light crudes, for example.

#### Viscosity

In some embodiments, the light cut may have a viscosity, measured at 40° C. according to ASTM D445, of less than 10 cSt. In other embodiments, the light cut may have a viscosity, measured at 40° C., of less than 5 cSt. In yet other embodiments, the light cut may have a viscosity, measured at 40° C., of less than 1 cSt. In some embodiments, the heavy cut may have a viscosity, measured at 100° C. according to ASTM D445, of greater than 10 cSt, such as greater than 20 cSt, greater than 35 cSt, greater than 50 cSt, greater than 75 cSt, or greater than 100 cSt. In various embodiments, the mid-cut may have a viscosity intermediate that of the light and heavy cuts.

Accordingly, in some embodiments, a light cut, including hydrocarbons having a boiling point up to about 300° C., for example, may have a viscosity, measured at 40° C., of less than 10 cSt; in other embodiments, such as when the light cut includes hydrocarbons having a boiling point up to about 250° C., for example, the light cut may have a viscosity, measured at 40° C., of less than 5 cSt; in yet other embodiments, such as when the light cut includes hydrocarbons having a boiling point up to about 220° C., for example, the

light cut may have a viscosity, measured at 40° C., of less than 1 cSt. In some embodiments where the light cut includes hydrocarbons having a boiling point below about 160° C., the light cut may have a viscosity, measured at 40° C., of less than 0.75 cSt. While the viscosity may vary for the different feeds at any given cut temperature, a low viscosity, such as less than 10 cSt, for example, has been found to improve the processability of the light hydrocarbons in the steam pyrolysis unit without the need for intermediate processing. Light cuts for Arab light crudes processed according to embodiments herein may target a viscosity of less than 0.55 cSt, for example, and may target a viscosity of less than 0.6 cSt for Arab extra light crudes, for example.

In various embodiments, the heavy cut, including hydrocarbons having a boiling point greater than about 300° C., may have a viscosity, measured at 100° C., of greater than 10 cSt. When the heavy cut includes hydrocarbons having a boiling point above about 350° C., the heavy cut may have a viscosity, measured at 100° C., of greater than 50 cSt. When the heavy cut includes hydrocarbons having a boiling point above about 400° C., the heavy cut may have a viscosity, measured at 100° C., of greater than 100 cSt. In embodiments where the heavy cut includes hydrocarbons having a boiling point above about 490° C., the heavy cut may have a viscosity of greater than 375 cSt, for example. A heavy-cut having a viscosity of greater than about 40 cSt, for example, has been found to be convertible to steam cracker feeds using the more severe destructive hydrogenation conditions in the heavy-cut conditioning section of processes herein.

#### MCRT

In some embodiments, the light cut may have only trace amounts, or undetectable amounts, of microcarbon residue (MCRT). In some embodiments, the mid cut may have a MCRT of less than 5 wt %, such as less than 3 wt %, less than 1 wt %, or less than 0.5 wt %. In some embodiments, the heavy cut may have an MCRT of greater than 0.5 wt %, such as greater than 1 wt %, greater than 3 wt %, greater than 5 wt %, or greater than 10 wt %.

In some embodiments, the mid cut, including hydrocarbons having a lower boiling point in the range from about 90° C. to about 300° C. and an upper boiling point in the range from about 400° C. to about 600° C., may have a MCRT of between about 0 wt % (trace or unmeasurable) and 1 wt %. A mid-cut having negligible MCRT, for example, has been found to be convertible to steam cracker feeds using relatively moderate destructive hydrogenation conditions in the mid-cut conditioning section of processes herein.

In various embodiments, the heavy cut, including hydrocarbons having a boiling point greater than about 300° C., may have a MCRT of greater than 0.5 wt %. When the heavy cut includes hydrocarbons having a boiling point above about 350° C., the heavy cut may have a MCRT of greater than 1 wt %. When the heavy cut includes hydrocarbons having a boiling point above about 400° C., the heavy cut may have a MCRT of greater than 5 wt %. In embodiments where the heavy cut includes hydrocarbons having a boiling point above about 490° C., the heavy cut may have a MCRT of greater than 15 wt %, for example. A heavy-cut having a MCCRT of greater than about 1 wt %, for example, has been found to be convertible to steam cracker feeds using the more severe destructive hydrogenation conditions in the heavy-cut conditioning section of processes herein.

#### Metals Content

In some embodiments, the light cut may have only trace amounts, or undetectable amounts, of metals. In some

embodiments, the mid cut may have a metals content of up to 50 ppm, such as less than 30 ppm, less than 10 ppm, or less than 1 ppm. In some embodiments, the heavy cut may have a metals content of greater than 1 ppm, such as greater than 10 ppm, greater than 20 ppm, greater than 35 ppm, or greater than 50 ppm.

In some embodiments, the mid cut, including hydrocarbons having a lower boiling point in the range from about 90° C. to about 300° C. and an upper boiling point in the range from about 400° C. to about 600° C., may have a metals content of between about 0 ppm (trace or unmeasurable) and 5 ppm, such as from greater than 0 ppm to 1 ppm. A mid-cut having negligible metals content, for example, has been found to be convertible to steam cracker feeds using relatively moderate destructive hydrogenation conditions in the mid-cut conditioning section of processes herein.

In various embodiments, the heavy cut, including hydrocarbons having a boiling point greater than about 300° C., may have a metals content of greater than 1 ppm. When the heavy cut includes hydrocarbons having a boiling point above about 350° C., the heavy cut may have a metals content of greater than 10 ppm. When the heavy cut includes hydrocarbons having a boiling point above about 400° C., the heavy cut may have a metals content of greater than 50 ppm. In embodiments where the heavy cut includes hydrocarbons having a boiling point above about 490° C., the heavy cut may have a metals content of greater than 75 ppm, for example. A heavy-cut having a metals content of greater than about 10 ppm, for example, has been found to be convertible to steam cracker feeds using the more severe destructive hydrogenation conditions in the heavy-cut conditioning section of processes herein.

As an example, an Arab Light crude oil stream may be separated in the initial separation step in order to produce the desired light-, mid-, and heavy-cuts. Without intending to be bound by theory, the light-cut may be a 160° C.- fraction with 5% of the fraction having a boiling point below 36° C. and 95% of the fraction having a boiling point below 160° C. (only 5% of the fraction would have a boiling point above 160° C.). The light cut may have an API gravity of about 65.5°, may have a BMCI of about 5.2, may have a hydrogen content of about 14.8 wt % (or 148,000 ppm), may have a nitrogen content of less than 0.00003 wt % (or 0.3 ppm), may have a sulfur content of about 0.0582 wt % (or 582 ppm), may have a viscosity, at 40° C., of about 0.5353 centistokes (cSt), and may have only trace amounts of MCRT and total metals content. The mid-cut may be a 160° C. to 490° C. fraction with 5% of the fraction having a boiling point below 173° C. and 95% of the fraction having a boiling point below 474° C. (only 5% of the fraction would have a boiling point above 474° C.). The mid-cut may have an API gravity of about 33.6°, may have a BMCI of about 25, may have a hydrogen content of about 12.83 wt % (or 128,300 ppm), may have a nitrogen content of less than 0.0227 wt % (or 227 ppm), may have a sulfur content of about 0.937 wt % (or 9,370 ppm), may have a viscosity, at 100° C., of about 1.58 centistokes (cSt), may have an MCRT of 0.03 wt %, and may have only trace amounts of total metals content. The heavy-cut may be a 490° C.+ fraction with 5% of the fraction having a boiling point below 490° C. and 95% of the fraction having a boiling point below 735° C. (only 5% of the fraction would have a boiling point above 735° C.). The heavy-cut may have an API gravity of about 8.2°, may have a BMCI of about 55, may have a hydrogen content of about 10.41 wt % (or 104,100 ppm), may have a nitrogen content of less than 0.2638 wt % (or 2,368 ppm), may have a sulfur content of about 3.9668 wt % (or 39,668

ppm), may have a viscosity, at 100° C., of about 394.3 centistokes (cSt), may have an MCRT of 17.22 wt %, and may have a total metals content 79.04 ppm.

As another example, an Arab Extra Light crude oil stream may be separated in the initial separation step in order to produce the desired light-, mid-, and heavy-cuts. Without intending to be bound by theory, the light-cut may be a 160° C.- fraction with 5% of the fraction having a boiling point below 54° C. and 95% of the fraction having a boiling point below 160° C. (only 5% of the fraction would have a boiling point above 160° C.). The light cut may have an API gravity of about 62°, may have a BMCI of about 9.09, may have a hydrogen content of about 14.53 wt % (or 145,300 ppm), may have a nitrogen content of less than 0.00003 wt % (or 0.3 ppm), may have a sulfur content of about 0.0472 wt % (or 472 ppm), may have a viscosity, at 40° C., of about 0.58 centistokes (cSt), and may have only trace amounts of MCRT and total metals content. The mid-cut may be a 160° C. to 490° C. fraction with 5% of the fraction having a boiling point below 169° C. and 95% of the fraction having a boiling point below 456° C. (only 5% of the fraction would have a boiling point above 474° C.). The mid-cut may have an API gravity of about 36.1°, may have a BMCI of about 21.22, may have a hydrogen content of about 13.38 wt % (or 133,800 ppm), may have a nitrogen content of less than 0.01322 wt % (or 132.2 ppm), may have a sulfur content of about 0.9047 wt % (or 9,047 ppm), may have a viscosity, at 100° C., of about 1.39 centistokes (cSt), and may have only trace amounts of MCRT and total metals content. The heavy-cut may be a 490° C.+ fraction with 5% of the fraction having a boiling point below 455° C. and 95% of the fraction having a boiling point below 735° C. (only 5% of the fraction would have a boiling point above 735° C.). The heavy-cut may have an API gravity of about 15.1°, may have a BMCI of about 33.28, may have a hydrogen content of about 11.45 wt % (or 114,500 ppm), may have a nitrogen content of less than 0.1599 wt % (or 1,599 ppm), may have a sulfur content of about 2.683 wt % (or 26,830 ppm), may have a viscosity, at 100° C., of about 48.79 centistokes (cSt), may have an MCRT of 9.53 wt %, and may have a total metals content 58.45 ppm.

While the various properties have been described with respect to Arab Light and Arab Extra Light, the aforementioned would also be applicable to other types of crude, such as desalted oils, condensate, biogenic oil, synthesis crude, tight oil, heavy hydrocarbons, reconstituted crudes, and bitumen derived oils.

Embodiments herein contemplate adjustment of the various cut points and reactor conditions based upon one or more of the above-noted properties. Methods according to embodiments herein may assay the petroleum feeds to be used, measuring one or more of the various properties of an incoming feed. Based on one or more of the properties, cut points, catalyst types (for moving bed reactors), pressures, temperatures, space velocity, hydrogen feed rates, and other variables may be adjusted to more effectively and efficiently utilize the reactor configuration, so as to maintain prime, near optimal, or optimal conditioning of the feedstock and the various cuts to desirable steam cracker feedstocks.

For example, the ebullated bed which receives the heavy-cut may have a capacity to process an amount of hydrocarbon having a sulfur content of less than 40,000 ppm. If a particular 490° C.+ heavy-cut would have a sulfur content of greater than 40,000 ppm, the capacity of the ebullated bed may be reduced. Accordingly, the heavy-cut point may be reduced, to 465° C.+, for example, in order have the sulfur content be less than 40,000 ppm. Further, if a particular 160°

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C.-490° C. mid-cut fraction has a hydrogen content of greater than 14 wt %, for example, and the nitrogen, sulfur, MCRT, and total metals is suitably low, the light-cut fraction may be expanded (from 160° C.- to 190° C.-, for example) to route more of the whole crude directly to the steam cracker. Alternatively, if the mid-cut is lower in hydrogen, for example, and/or the sulfur, nitrogen, MCRT, and/or total metals are not suitably low, the light-cut may be reduced (from 160° C.- to 130° C.-, for example), such that additional mid-cut may be processed in the fixed bed conditioning stages.

As described above, embodiments herein may relate to one or more of the following embodiments:

Embodiment 1: A process for converting whole crudes and other heavy hydrocarbon streams to produce olefins and/or aromatics, the process comprising:

feeding a whole crude into a heater, producing a pre-heated hydrocarbon feedstock;

separating the pre-heated hydrocarbon feedstock in a separator into a light boiling fraction and an intermediate fraction;

feeding the intermediate fraction back to the heater, producing a heated intermediate fraction;

feeding a hydrogen stream to a hot hydrogen stripper;

separating the heated intermediate fraction in the hot hydrogen stripper into a medium boiling fraction and a hot hydrogen stripper bottoms fraction; and

cooling the hot hydrogen stripper bottoms fraction via indirect heat exchange against the intermediate fraction producing a high boiling residue fraction;

separating the high boiling fraction in a solvent deasphalting unit to produce a deasphalted oil fraction and a pitch fraction;

destructively hydrogenating the medium boiling fraction and the deasphalted oil fraction to produce a hydrotreated effluent;

feeding the hydrotreated effluent and the light boiling fraction to a steam cracker to convert hydrocarbons therein into one or more light olefins and a pyrolysis oil.

Embodiment 2: The process of embodiment 1, wherein the light boiling fraction has two or more of the following properties:

a 95% boiling point temperature in the range from about 130° C. to about 200° C.;

a hydrogen content of at least 14 wt %;

a BMCI of less than 5;

an API gravity of greater than 40°;

a sulfur content of less than 1000 ppm;

a nitrogen content of less than 10 ppm;

a viscosity, measured at 40° C., of less than 1 cSt; less than 1 wt % MCRT; and

less than 1 ppm total metals.

Embodiment 3: The process of embodiment 1 or embodiment 2, wherein the medium boiling fraction has two or more of the following properties:

a 5% boiling point temperature in the range from about 130° C. to about 200° C.;

a 95% boiling point temperature in the range from about 400° C. to about 600° C.;

a hydrogen content in the range from about 12 wt % to about 14 wt %;

a BMCI in the range from about 5 to less than 50;

an API gravity of in the range from about 10° to about 40°;

a sulfur content in the range from about 1000 ppm to about 10000 ppm;

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a nitrogen content in the range from about 1 ppm to about 100 ppm;

a viscosity, measured at 40° C., of greater than 1 cSt;

less than 5 wt % MCRT; and

less than 50 ppm total metals.

Embodiment 4: The process of any one of embodiments 1-3, wherein the high boiling residue fraction has two or more of the following properties:

a 5% boiling point temperature in the range from about 400° C. to about 600° C.;

a hydrogen content of less than 12 wt %;

a BMCI of greater than 50;

an API gravity of less than 10°;

a sulfur content of greater than 10000 ppm;

a nitrogen content of greater than 100 ppm;

a viscosity, measured at 100° C., of greater than 100 cSt; greater than 5 wt % MCRT; and

greater than 50 ppm total metals.

Embodiment 5: The process of any one of embodiments 1-4, wherein an overall chemicals production of the feedstock is at least 65 wt %, based on the total amount of olefins produced as compared to a total feedstock feed rate.

Embodiment 6: The process of any one of embodiments 1-5, wherein destructively hydrogenating comprises converting hydrocarbons in the high boiling residue fraction to one or more naphtha range hydrocarbons.

Embodiment 7: The process of embodiment 2, wherein the light boiling fraction does not comprise hydrocarbons having a boiling point of greater than 160° C.

Embodiment 8: The process of embodiment 1, further comprising recycling the pyrolysis oil to the solvent deasphalting unit.

Embodiment 9: The process of embodiment 1, wherein the destructive hydrogenation of the medium boiling fraction and the deasphalted oil fraction comprises:

destructively hydrogenation the medium boiling fraction in a first hydroprocessing unit;

destructively hydrogenation the deasphalted oil fraction in a second hydroprocessing unit.

Embodiment 10: The process of embodiment 9, wherein destructively hydrogenation the medium boiling fraction comprises converting hydrocarbons in the medium boiling fraction to primarily naphtha range hydrocarbons.

Embodiment 11: The process of embodiment 9, wherein destructively hydrogenation the deasphalted oil fraction comprises converting hydrocarbons in the deasphalted oil fraction to primarily naphtha range hydrocarbons.

Embodiment 12: A system for converting whole crudes and other heavy hydrocarbon streams to produce olefins, the system comprising:

a separation unit for separating a whole crude into at least a light boiling fraction, a medium boiling fraction, and a high boiling residue fraction;

a solvent deasphalting unit for deasphalting the high boiling residue fraction and producing a deasphalted oil fraction and a pitch fraction;

a conditioning system for destructively hydrogenating the medium boiling fraction and the deasphalted fraction to produce a hydrotreated effluent;

a steam cracker unit for converting the hydrotreated effluent and the light boiling fraction into one or more light olefins and a pyrolysis oil.

Embodiment 13: The system of embodiment 12, wherein the light boiling fraction has two or more of the following properties:

a 95% boiling point temperature in the range from about 130° C. to about 200° C.;

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a hydrogen content of at least 14 wt %;  
 a BMCI of less than 5;  
 an API gravity of greater than 40°;  
 a sulfur content of less than 1000 ppm;  
 a nitrogen content of less than 10 ppm;  
 a viscosity, measured at 40° C., of less than 1 cSt;  
 less than 1 wt % MCRT; and  
 less than 1 ppm total metals.

Embodiment 14: The system of embodiments 12 or  
 embodiment 13, wherein the medium boiling fraction has  
 two or more of the following properties:

a 5% boiling point temperature in the range from about  
 130° C. to about 200° C.;  
 a 95% boiling point temperature in the range from about  
 400° C. to about 600° C.;  
 a hydrogen content in the range from about 12 wt % to  
 about 14 wt %;  
 a BMCI in the range from about 5 to less than 50;  
 an API gravity of in the range from about 10° to about  
 40°;  
 a sulfur content in the range from about 1000 ppm to  
 about 10000 ppm;  
 a nitrogen content in the range from about 1 ppm to about  
 100 ppm;  
 a viscosity, measured at 40° C., of greater than 1 cSt;  
 less than 5 wt % MCRT; and  
 less than 50 ppm total metals.

Embodiment 15: The system of any one of embodiments  
 12-14, wherein the high boiling residue fraction has two or  
 more of the following properties:

a 5% boiling point temperature in the range from about  
 400° C. to about 600° C.;  
 a hydrogen content of less than 12 wt %;  
 a BMCI of greater than 50;  
 an API gravity of less than 10°;  
 a sulfur content of greater than 10000 ppm;  
 a nitrogen content of greater than 100 ppm;  
 a viscosity, measured at 100° C., of greater than 100 cSt;  
 greater than 5 wt % MCRT; and  
 greater than 50 ppm total metals.

Embodiment 16: The system of embodiment 12, wherein  
 the conditioning system for hydrotreating the medium boiling  
 fraction and the deasphalted oil fraction comprises:

a first hydroprocessing unit for hydroprocessing the  
 medium boiling fraction; and  
 a second hydroprocessing unit for hydroprocessing the  
 deasphalted oil fraction; and

Embodiment 17: The system of embodiment 12, further  
 comprising a flow line for diverting the deasphalted oil  
 fraction to the first hydroprocessing unit.

Embodiment 18: The system of embodiment 16, wherein  
 the first hydroprocessing unit comprises catalyst configured  
 for converting hydrocarbons in the medium boiling fraction  
 to primarily naphtha range hydrocarbons.

Embodiment 19: The system of embodiment 16, wherein  
 the second hydroprocessing unit comprises catalyst config-  
 ured for converting hydrocarbons in the deasphalted oil  
 fraction to primarily naphtha range hydrocarbons.

Embodiment 20: The system of embodiment 12, further  
 comprising a flow line configured for recycling the pyrolysis  
 oil to the solvent deasphalting unit.

Embodiment 21: A system for the thermal conversion of  
 crude oils to chemicals, the system comprising:

a conditioning system comprising one or more hydrotreat-  
 ing and/or hydrocracking reactors configured to condi-  
 tion a whole crude oil for use as a steam cracker  
 feedstock;

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a steam cracker for converting the steam cracker feed-  
 stock into chemicals including ethylene, propylene, and  
 butenes;

wherein the conditioning unit and the steam cracker are  
 configured to have aligned run times.

Embodiment 22: A process for the thermal conversion of  
 crude oils to chemicals, the system comprising:

hydrotreating and/or hydrocracking a whole crude in a  
 conditioning unit comprising one or more reactors;  
 feeding the hydrotreated and/or hydrocracked whole  
 crude to a steam cracker for converting the  
 hydrotreated and/or hydrocracked whole crude into  
 chemicals including ethylene, propylene, and butenes;  
 wherein the conditioning unit and the steam cracker are  
 configured to have aligned run times.

Embodiment 23: The process of embodiment 19, wherein  
 the whole crude is a condensate hydrocarbon with a boiling  
 range up to 565° C.

Embodiment 24: A process for converting whole crudes  
 and other heavy hydrocarbon streams to produce olefins  
 and/or aromatics, the process comprising:

separating a whole crude into at least light boiling frac-  
 tion, a medium boiling fraction, and a high boiling  
 residue fraction;

separating the high boiling residue into a pitch fraction  
 and a deasphalted oil fraction;

destructively dehydrogenating the deasphalted oil fraction  
 in a hydroprocessing system to produce a hydrotreated  
 effluent;

destructively hydrogenating the medium boiling fraction  
 in a second hydroprocessing system to produce a steam  
 cracker feed;

feeding the light boiling fraction, the steam cracker feed,  
 and the hydrotreated effluent to a steam cracker to  
 convert hydrocarbons therein into one or more light  
 olefins and a pyrolysis oil.

Embodiment 25: The process of embodiment 24, wherein  
 the light boiling fraction has two or more of the following  
 properties:

a 95% boiling point temperature in the range from about  
 130° C. to about 200° C.;

a hydrogen content of at least 14 wt %;

a BMCI of less than 5;

an API gravity of greater than 40°;

a sulfur content of less than 1000 ppm;

a nitrogen content of less than 10 ppm;

a viscosity, measured at 40° C., of less than 1 cSt;

less than 1 wt % MCRT; and

less than 1 ppm total metals.

Embodiment 26: The process of embodiment 24 or  
 embodiment 25, wherein the medium boiling fraction has  
 two or more of the following properties:

a 5% boiling point temperature in the range from about  
 130° C. to about 200° C.;

a 95% boiling point temperature in the range from about  
 400° C. to about 600° C.;

a hydrogen content in the range from about 12 wt % to  
 about 14 wt %;

a BMCI in the range from about 5 to less than 50;

an API gravity of in the range from about 10° to about  
 40°;

a sulfur content in the range from about 1000 ppm to  
 about 10000 ppm;

a nitrogen content in the range from about 1 ppm to about  
 100 ppm;

a viscosity, measured at 40° C., of greater than 1 cSt;  
 less than 5 wt % MCRT; and

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less than 50 ppm total metals.

Embodiment 27: The process of any one of embodiments 24-26, wherein the high boiling residue fraction has two or more of the following properties:

a 5% boiling point temperature in the range from about 5 400° C.-600° C.;

a hydrogen content of less than 12 wt %;

a BMCI of greater than 50;

an API gravity of less than 10°;

a sulfur content of greater than 10000 ppm;

a nitrogen content of greater than 100 ppm;

a viscosity, measured at 100° C., of greater than 100 cSt; greater than 5 wt % MCRT; and greater than 50 ppm total metals.

Embodiment 28: The process of any one of embodiments 24-27, wherein an overall chemicals production of the feedstock is at least 65 wt %, based on the total amount of olefins produced as compared to a total feedstock feed rate.

Embodiment 29: The process of any one of embodiments 24-28, wherein destructively hydrogenating the high boiling residue fraction comprises converting hydrocarbons in the high boiling residue fraction to one or more naphtha range hydrocarbons.

Embodiment 30: the process of embodiment 24, wherein the separating the whole crude comprises

feeding the whole crude into a heater, producing a pre-heated hydrocarbon feedstock;

separating the pre-heated hydrocarbon feedstock in a separator into the light boiling fraction and an intermediate fraction;

feeding the intermediate fraction back to the heater, producing a heated intermediate fraction;

feeding a hydrogen stream and the intermediate fraction to a hot hydrogen stripper;

separating the heated intermediate fraction in the hot hydrogen stripper into the medium boiling fraction and a hot hydrogen stripper bottoms fraction

cooling the hot hydrogen stripper bottoms fraction via indirect heat exchange against the intermediate fraction producing the high boiling residue fraction.

Embodiment 31: A process for converting whole crudes and other heavy hydrocarbon streams to produce olefins and/or aromatics, the process comprising:

separating a whole crude into at least a light boiling fraction, a medium boiling fraction, and a high boiling residue fraction;

separating the high boiling fraction in a solvent deasphalting unit to produce a pitch fraction and a deasphalted oil fraction;

destructively hydrogenating the deasphalted oil fraction and a pyrolysis oil in a hydroprocessing system to produce a hydroprocessed fraction;

destructively hydrogenating the medium boiling fraction in a second hydroprocessing system to produce a steam cracker feed;

feeding the light boiling fraction, the steam cracker feed, and the hydrotreated effluent to a steam cracker to convert hydrocarbons therein into one or more light olefins and the pyrolysis oil;

recycling the pyrolysis oil to the solvent deasphalting unit.

Embodiment 32: The process of embodiment 31, wherein the light boiling fraction has two or more of the following properties:

a 95% boiling point temperature in the range from about 65 130° C. to about 200° C.;

a hydrogen content of at least 14 wt %;

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a BMCI of less than 5;

an API gravity of greater than 40°;

a sulfur content of less than 1000 ppm;

a nitrogen content of less than 10 ppm;

a viscosity, measured at 40° C., of less than 1 cSt;

less than 1 wt % MCRT; and

less than 1 ppm total metals.

Embodiment 33: The process of embodiment 31 or embodiment 32, wherein the medium boiling fraction has two or more of the following properties:

a 5% boiling point temperature in the range from about 130° C. to about 200° C.;

a 95% boiling point temperature in the range from about 400° C. to about 600° C.;

a hydrogen content in the range from about 12 wt % to about 14 wt %;

a BMCI in the range from about 5 to less than 50;

an API gravity of in the range from about 10° to about 40°;

a sulfur content in the range from about 1000 ppm to about 10000 ppm;

a nitrogen content in the range from about 1 ppm to about 100 ppm;

a viscosity, measured at 40° C., of greater than 1 cSt;

less than 5 wt % MCRT; and

less than 50 ppm total metals.

Embodiment 34: The process of any one of embodiments 31-33, wherein the high boiling residue fraction has two or more of the following properties:

a 5% boiling point temperature in the range from about 400° C.-600° C.;

a hydrogen content of less than 12 wt %;

a BMCI of greater than 50;

an API gravity of less than 10°;

a sulfur content of greater than 10000 ppm;

a nitrogen content of greater than 100 ppm;

a viscosity, measured at 100° C., of greater than 100 cSt; greater than 5 wt % MCRT; and greater than 50 ppm total metals.

Embodiment 35: The process of any one of embodiments 31-34, wherein an overall chemicals production of the feedstock is at least 65 wt %, based on the total amount of olefins produced as compared to a total feedstock feed rate.

Embodiment 36: The process of any one of embodiments 31-35, wherein destructively hydrogenating the deasphalted oil fraction comprises converting hydrocarbons in the high boiling residue fraction to one or more naphtha range hydrocarbons.

Embodiment 37: the process of embodiment 31, wherein the separating the whole crude comprises

feeding the whole crude into a heater, producing a pre-heated hydrocarbon feedstock;

separating the pre-heated hydrocarbon feedstock in a separator into the light boiling fraction and an intermediate fraction;

feeding the intermediate fraction back to the heater, producing a heated intermediate fraction;

feeding a hydrogen stream and the intermediate fraction to a hot hydrogen stripper;

separating the heated intermediate fraction in the hot hydrogen stripper into the medium boiling fraction and a hot hydrogen stripper bottoms fraction

cooling the hot hydrogen stripper bottoms fraction via indirect heat exchange against the intermediate fraction producing the high boiling residue fraction.

While the disclosure includes a limited number of embodiments, those skilled in the art, having benefit of this

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disclosure, will appreciate that other embodiments may be devised which do not depart from the scope of the present disclosure.

We claim:

1. A system for converting whole crudes and other heavy hydrocarbon streams to produce olefins, the system comprising:

- a separation unit for separating a whole crude into at least a light boiling fraction, a medium boiling fraction, and a high boiling residue fraction;
- a solvent deasphalting unit for deasphalting the high boiling residue fraction and producing a deasphalted oil fraction and a pitch fraction;
- a conditioning system for destructively hydrogenating the medium boiling fraction and the deasphalted fraction to produce a hydrotreated effluent;
- a steam cracker unit for converting the hydrotreated effluent and the light boiling fraction into one or more light olefins and a pyrolysis oil.

2. The system of claim 1, wherein the conditioning system for hydrotreating the medium boiling fraction and the deasphalted oil fraction comprises:

- a first hydroprocessing unit for hydroprocessing the medium boiling fraction; and
- a second hydroprocessing unit for hydroprocessing the deasphalted oil fraction.

3. The system of claim 2, further comprising a flow line for diverting the deasphalted oil fraction to the first hydroprocessing unit.

4. The system of claim 2, wherein the first hydroprocessing unit comprises catalyst configured for converting hydrocarbons in the medium boiling fraction to primarily steam crackable products.

5. The system of claim 2, wherein the second hydroprocessing unit comprises catalyst configured for converting hydrocarbons in the deasphalted oil fraction to primarily steam crackable products.

6. The system of claim 1, further comprising a flow line configured for recycling the pyrolysis oil to the solvent deasphalting unit.

7. A system for the thermal conversion of crude oils to chemicals, the system comprising:

- a conditioning system comprising one or more hydrotreating and/or hydrocracking reactors configured to condition a whole crude oil for use as a steam cracker feedstock;
- a steam cracker for converting the steam cracker feedstock into chemicals including ethylene, propylene, and butenes;

wherein the conditioning unit and the steam cracker are configured to have aligned run times.

8. The system of claim 7, wherein the conditioning system comprises:

- a first hydrotreating and/or hydrocracking unit for hydrotreating and/or hydrocracking a medium boiling fraction; and

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a second hydrotreating and/or hydrocracking unit for hydrotreating and/or hydrocracking a deasphalted oil fraction.

9. The system of claim 8, further comprising a flow line for diverting the deasphalted oil fraction to the first hydrotreating and/or hydrocracking unit.

10. The system of claim 8, wherein the first hydroprocessing unit comprises catalyst configured for converting hydrocarbons in the medium boiling fraction to primarily steam crackable products.

11. The system of claim 8, wherein the second hydroprocessing unit comprises catalyst configured for converting hydrocarbons in the deasphalted oil fraction to primarily steam crackable products.

12. The system of claim 8, further comprising a flow line configured for recycling the pyrolysis oil to the solvent deasphalting unit.

13. A system for converting whole crudes and other heavy hydrocarbon streams to produce olefins, the system comprising:

- a separation unit for separating a whole crude into at least a medium boiling fraction and a high boiling residue fraction;

a solvent deasphalting unit for deasphalting the high boiling residue fraction and producing a deasphalted oil fraction and a pitch fraction;

a conditioning system for destructively hydrogenating the medium boiling fraction and the deasphalted fraction to produce a hydrotreated effluent;

a steam cracker unit for converting the hydrotreated effluent into one or more light olefins and a pyrolysis oil.

14. The system of claim 13, wherein the conditioning system for hydrotreating the medium boiling fraction and the deasphalted oil fraction comprises:

- a first hydroprocessing unit for hydroprocessing the medium boiling fraction; and

a second hydroprocessing unit for hydroprocessing the deasphalted oil fraction.

15. The system of claim 14, further comprising a flow line for diverting the deasphalted oil fraction to the first hydroprocessing unit.

16. The system of claim 14, wherein the first hydroprocessing unit comprises catalyst configured for converting hydrocarbons in the medium boiling fraction to primarily steam crackable products.

17. The system of claim 14, wherein the second hydroprocessing unit comprises catalyst configured for converting hydrocarbons in the deasphalted oil fraction to primarily steam crackable products.

18. The system of claim 13, further comprising a flow line configured for recycling the pyrolysis oil to the solvent deasphalting unit.

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